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IMS566

ADVANCED WEB DESIGN DEVELOPMENT AND CONTENT MANAGEMENT

SPORTS FACILITY BOOKING MANAGEMENT SYSTEM USING LARAVEL FRAMEWORK

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ABSTRACT

The Sports Facility Booking Management System is a web-based application developed to improve the efficiency of managing sports facility reservations at Universiti Teknologi MARA (UiTM). Traditional booking methods often involve manual processes that are time-consuming, prone to scheduling conflicts, and difficult to monitor effectively. This project aims to provide a centralized online platform that simplifies the reservation process for students while enabling administrators to manage booking activities efficiently.

The system was developed using the Laravel Framework following the Model-View-Controller (MVC) architecture. MySQL was used as the database management system to store user, facility, and booking information securely. Bootstrap was implemented to provide a responsive and user-friendly interface, while FullCalendar was integrated to visualize booking schedules. Chart.js was also utilized to display booking statistics through interactive charts on the dashboard.

The developed system consists of two user roles: Administrator and Student. Students are able to register, log in, browse available sports facilities, submit booking requests, view booking history, and manage their profiles. Administrators are responsible for managing facilities, facility types, facility features, operating hours, booking approvals, reports, calendars, and overall system administration.

The implementation of this system successfully reduces manual workload, minimizes booking conflicts, improves record management, and enhances the overall efficiency of sports facility reservation processes. The project demonstrates how modern web technologies can be applied to develop an efficient, secure, and user-friendly booking management system suitable for educational institutions.

Keywords: Laravel Framework, Sports Facility Booking, Web Application, MySQL, Information System, Reservation System.

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INTRODUCTION

1.1 Introduction

The rapid advancement of information technology has significantly transformed the way organizations manage daily operations and provide services to users. Web-based management systems have become essential in improving efficiency, minimizing human errors, and providing users with faster access to services. Educational institutions are increasingly adopting online systems to automate administrative processes, allowing students and staff to perform various activities conveniently without relying on manual procedures.

Sports facilities play an important role in supporting students' physical development, recreational activities, and co-curricular programmes. At Universiti Teknologi MARA (UiTM), various sports facilities such as badminton courts, futsal courts, basketball courts, volleyball courts, tennis courts, and multipurpose halls are provided for students and staff. However, the booking process is often performed manually through physical forms, telephone calls, or direct communication with administrators. Such practices frequently result in scheduling conflicts, double bookings, inefficient record management, and delays in confirming reservations.

To overcome these challenges, this project proposes the development of a web-based **Sports Facility Booking Management System** using the Laravel Framework. The system provides an integrated platform that enables students to submit booking requests online while allowing administrators to manage facilities, approve reservations, monitor booking activities, and generate reports efficiently. Through the implementation of modern web technologies, the proposed system aims to improve operational efficiency, enhance user satisfaction, and provide a more organized booking management process.

1.2 Background of Study

Sports facilities are valuable assets within higher learning institutions as they encourage healthy lifestyles, support university sports programmes, and provide spaces for recreational activities. Due to the increasing number of students utilizing these facilities, an efficient booking management system is required to ensure fair usage and proper scheduling.

Traditionally, facility reservations are managed manually, requiring students to visit the administration office or contact staff to check facility availability. This manual process consumes considerable time for both students and administrators. Furthermore, maintaining paper records or manually updating spreadsheets increases the likelihood of human errors, misplaced records, and overlapping reservations.

The rapid growth of web technologies has enabled organizations to digitalize many administrative processes. Laravel Framework is one of the most popular PHP frameworks due to its Model-View-Controller (MVC) architecture, security features, routing capabilities, and database management support. Combined with MySQL, Bootstrap, FullCalendar, and Chart.js, Laravel provides a robust platform for developing a modern web-based booking management system.

Therefore, this project focuses on designing and developing a Sports Facility Booking Management System that automates the booking process, improves facility management, and provides users with a secure and user-friendly online reservation platform.

1.3 Problem Statement

The current sports facility booking process at many educational institutions still relies on manual methods that create various operational challenges. Students often experience difficulties in checking facility availability, while administrators spend considerable time processing booking requests and maintaining records. These issues reduce operational efficiency and increase the possibility of scheduling conflicts.

The major problems identified are as follows:

1. Students are unable to determine real-time facility availability before submitting booking requests.
2. Manual booking procedures increase the possibility of double-booking and scheduling conflicts.
3. Administrators require significant time to process booking requests, update booking records, and manage facility schedules manually.
4. Booking reports and reservation statistics cannot be generated automatically, making monitoring and decision-making more difficult.
5. Existing manual processes increase the risk of data loss, inconsistent records, and human errors.

These problems highlight the need for a computerized booking management system capable of automating reservation processes while providing a centralized database for efficient information management.

1.4 Project Objectives

The objectives of this project are:

General Objective

To develop a web-based Sports Facility Booking Management System using Laravel Framework for managing sports facility reservations at Universiti Teknologi MARA.

Specific Objectives

1. To develop an online booking system that allows students to reserve sports facilities conveniently.
2. To provide administrators with an efficient platform to manage facilities, booking requests, and user information.
3. To minimize booking conflicts through automated reservation validation and approval mechanisms.
4. To provide administrators with an efficient booking management system that supports booking approval, rejection, completion, and scheduling conflict prevention.
5. To provide a responsive, secure, and user-friendly web application accessible through modern web browsers.

1.5 Project Scope

The proposed system is designed specifically for managing sports facility reservations within Universiti Teknologi MARA. The project consists of two categories of users: administrators and students. Each user is provided with different system functionalities according to their responsibilities.

Administrator Scope

The administrator is responsible for managing the overall operation of the booking system. The administrator can:

- Manage student accounts
- Manage sports facilities
- Manage facility types
- Manage facility features
- Manage operating hours
- Approve booking requests
- Reject booking requests
- Complete booking records
- View booking calendar
- Manage personal profile

Student Scope

Students are able to access the system after authentication and can perform the following functions:

- Register an account
- Login securely
- View dashboard
- Browse available facilities
- Submit booking requests
- View booking history
- Cancel pending bookings
- View booking calendar
- Update personal profile

The project is developed as a web-based application using Laravel Framework and MySQL database. Mobile application development and online payment integration are outside the scope of this project.

1.6 Significance of the Project

The implementation of the Sports Facility Booking Management System provides significant benefits to administrators, students, and the university.

Benefits to Students

Students are able to submit booking requests anytime through an online platform without visiting the administration office. They can monitor booking status, review booking history, and receive faster responses from administrators.

Benefits to Administrators

The system simplifies administrative tasks by automating booking approvals, maintaining centralized booking records and monitoring facility usage through dashboards and calendar visualization.

Benefits to Universiti Teknologi MARA

The university benefits from improved facility utilization, reduced administrative workload, minimized booking conflicts, and enhanced service quality provided to students.

1.7 Project Limitation

Although the proposed system successfully automates the sports facility booking process, several limitations exist.

Firstly, the system is developed specifically for web browsers and does not include a mobile application version. Secondly, the system does not integrate online payment services because all facility bookings are assumed to be free of charge. Thirdly, email and SMS notifications are not implemented within the current version. In addition, the system is designed for use within Universiti Teknologi MARA only and may require further customization before deployment in other organizations.

1.8 Report Organization

This report is organized into five chapters.

Chapter One introduces the project background, problem statement, objectives, scope, significance, limitations, and overall report structure.

Chapter Two presents the literature review, including previous studies related to online booking systems, Laravel Framework, Bootstrap Framework, MySQL database, FullCalendar, Chart.js, and Model-View-Controller architecture.

Chapter Three explains the research methodology, software development life cycle, requirement analysis, system architecture, database design, use case diagrams, activity diagrams, sequence diagrams, entity relationship diagrams, and user interface design.

Chapter Four discusses the implementation of the Sports Facility Booking Management System. This chapter explains each system module, including authentication, dashboards, booking management, facility management, booking conflicts validation, calendar, profile management, and system testing.

Chapter Five concludes the project by summarizing the project achievements, discussing system limitations, proposing future enhancements, and providing recommendations for further development.

LITERATURE REVIEW

2.1 Introduction

This chapter presents the literature review related to the development of the Sports Facility Booking Management System. It discusses the concepts, technologies, and previous studies that support the development of the proposed system. The review includes web-based information systems, sports facility booking systems, Laravel Framework, Model-View-Controller (MVC) architecture, MySQL Database Management System, Bootstrap Framework, FullCalendar, and Chart.js. In addition, a comparison of existing systems is conducted to identify useful features that can be implemented in the proposed system.

The purpose of this chapter is to provide a theoretical foundation for the project and justify the selection of technologies used throughout the system development process. Understanding these concepts ensures that the proposed system is developed using appropriate methods and modern web development practices.

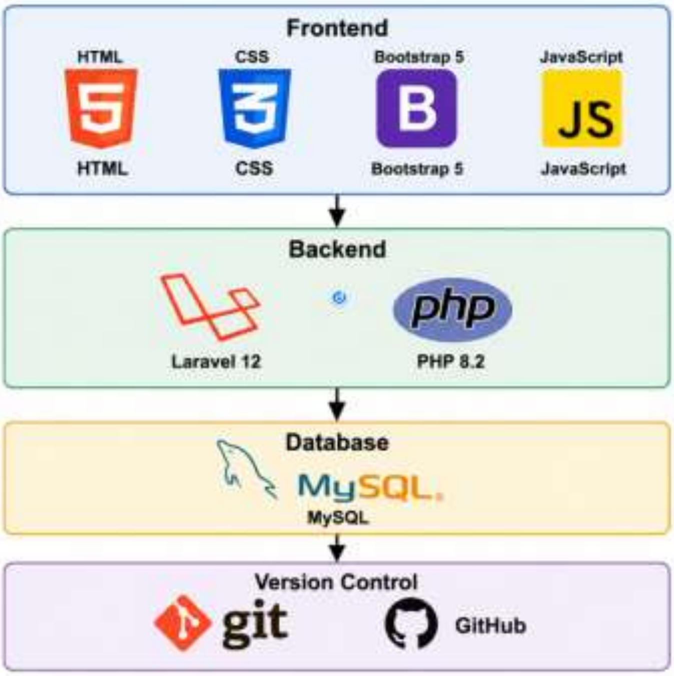

Figure 2.1 Technology Stack of the Proposed Sports Facility Booking Management System

2.2 Sports Facility Booking System

A Sports Facility Booking System is an information system developed to simplify the reservation process of sports facilities by allowing users to submit booking requests electronically. Compared to manual booking methods, an online booking system enables users to check facility availability, make reservations, and receive booking updates through a centralized platform. This approach reduces paperwork, improves scheduling efficiency, and minimizes booking conflicts.

Many educational institutions have adopted online booking systems to improve the management of sports facilities. These systems provide administrators with better control over facility usage while allowing students to make reservations more conveniently. Features such as booking approval, booking history, availability checking, and report generation have become essential components of modern reservation systems.

The proposed Sports Facility Booking Management System incorporates these features by providing two categories of users: administrators and students. Students are able to browse facilities, submit booking requests, and monitor reservation status, while administrators manage facilities, approve bookings, reject bookings, complete bookings, monitor facility schedules, and manage overall booking activities efficiently.

Figure 2.2 Homepage or Dashboard of the Proposed System

2.3 Web-Based Information System

A web-based information system is an application that operates through a web browser using internet or local network connectivity. Unlike desktop applications, web-based systems do not require installation on individual computers, allowing users to access the system from different devices at any time. This flexibility makes web applications suitable for organizations that require centralized information management.

The Sports Facility Booking Management System was developed as a web-based application to provide convenient access for both students and administrators. Users can access the system using computers, laptops, tablets, or smartphones without installing additional software. All booking information is stored centrally within a MySQL database, ensuring data consistency and simplifying maintenance.

The implementation of a web-based platform also improves communication between users and administrators by allowing booking requests and approvals to be processed online. This significantly reduces the time required to complete the reservation process compared to traditional manual methods.

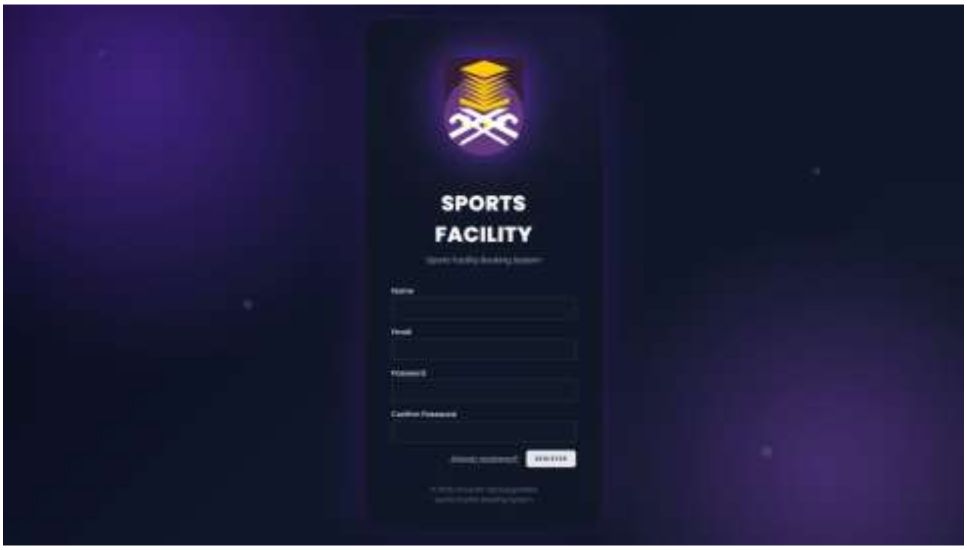

Figure 2.3 Login Interface of the Proposed System

2.4 Laravel Framework

Laravel is an open-source PHP framework that follows the Model-View-Controller (MVC) architectural pattern. It provides various built-in features such as authentication, routing, middleware, database migration, validation, and Object Relational Mapping (ORM) through Eloquent. These features simplify application development while improving security and maintainability.

Laravel was selected as the primary development framework for this project because it provides a structured coding environment and supports rapid application development. The framework enables developers to organize source code efficiently by separating application logic into models, controllers, and views.

In this project, Laravel is used to manage user authentication, process booking requests, interact with the MySQL database, process booking requests, validate booking conflicts, manage user authentication and control access permissions. Laravel Breeze is also implemented to provide secure login and registration functionalities.

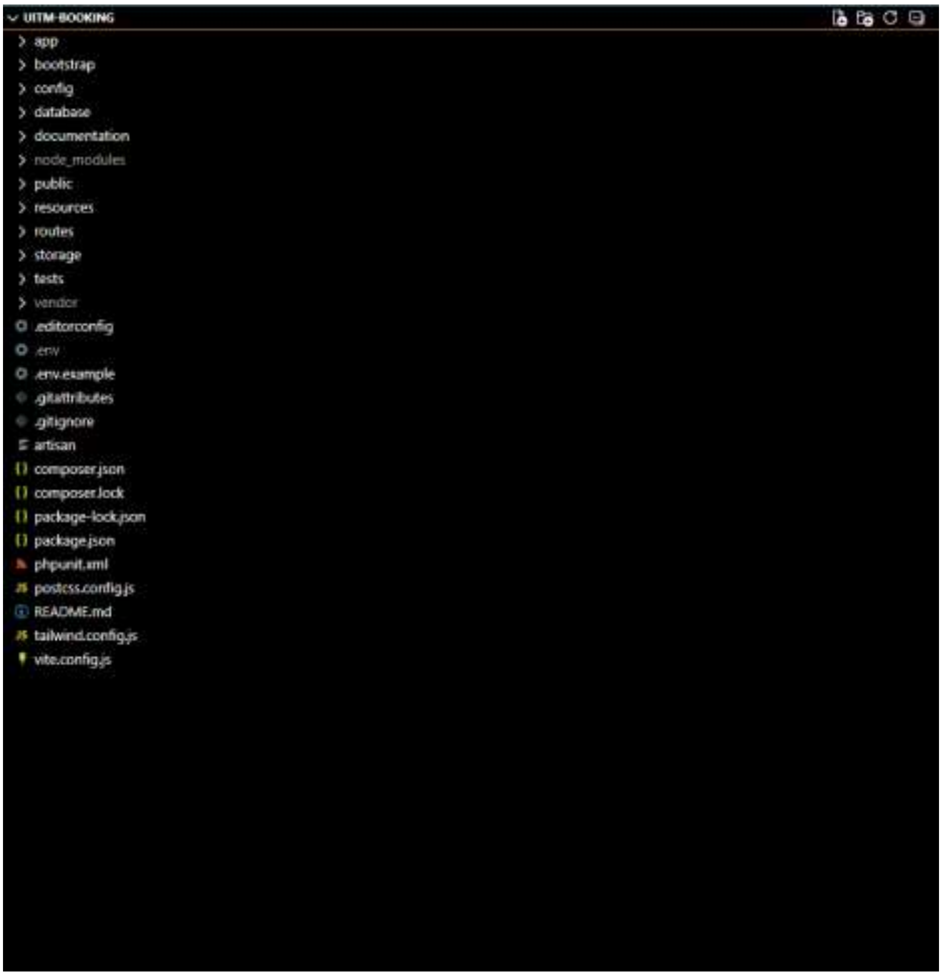

Figure 2.4 Laravel Project Structure in Visual Studio Code

2.5 Model-View-Controller (MVC) Architecture

The Model-View-Controller (MVC) architecture separates an application into three interconnected components: Model, View, and Controller. This design pattern improves system organization by separating business logic, data processing, and user interface components.

The Model represents the application's data and communicates directly with the database. The View displays information to users through graphical interfaces, while the Controller processes user requests and coordinates communication between the Model and the View.

Laravel fully implements the MVC architecture, making it easier to maintain and expand the application. Within the proposed system, controllers handle booking operations, models interact with database tables, and Blade templates generate responsive user interfaces for administrators and students.

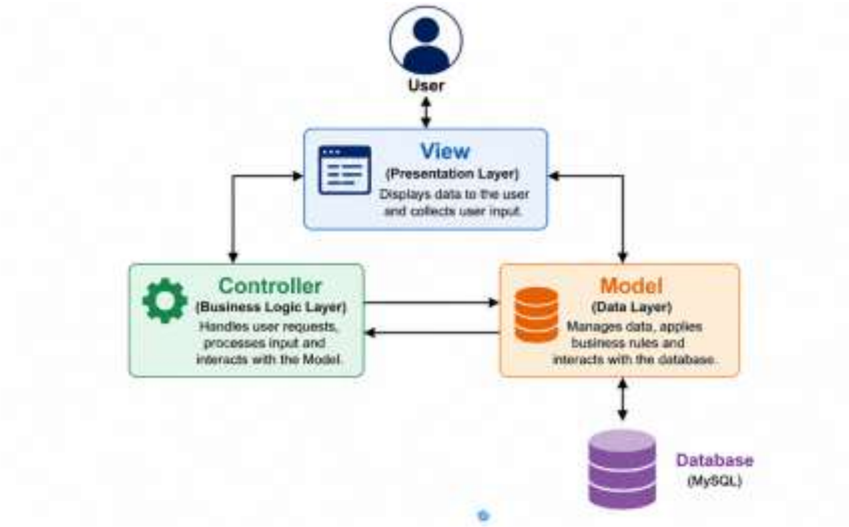

Figure 2.5 Model-View-Controller (MVC) Architecture

2.6 MySQL Database Management System

MySQL is a relational database management system widely used for storing structured information within web applications. It organizes data into related tables using primary keys and foreign keys to maintain data integrity and reduce redundancy.

The Sports Facility Booking Management System uses MySQL to manage user accounts, facility information, booking records, facility types, facility features, and operating hours. The relational database structure enables efficient retrieval of booking information while ensuring consistency across different system modules.

MySQL was selected because of its reliability, compatibility with Laravel, and ability to support multi-user environments efficiently.

Table Name	Description
Users	Stores administrator and student information
Facilities	Stores sports facility information
Facility Types	Stores categories of facilities
Facility Features	Stores available facility features
Operating Hours	Stores operating schedules
Bookings	Stores reservation records

Table 2.1 Database Tables Used in the Proposed System

2.7 Bootstrap Framework

Bootstrap is an open-source front-end framework that provides responsive interface components for modern web applications. It includes predefined layouts, navigation bars, forms, tables, buttons, and cards that simplify interface development while ensuring consistency across different devices.

Bootstrap was implemented throughout the Sports Facility Booking Management System to create responsive dashboards, booking forms, management pages, and reports. The framework ensures that the application remains user-friendly and visually consistent regardless of screen size.

The responsive design provided by Bootstrap allows students and administrators to access the system using desktops, tablets, and smartphones without affecting usability.

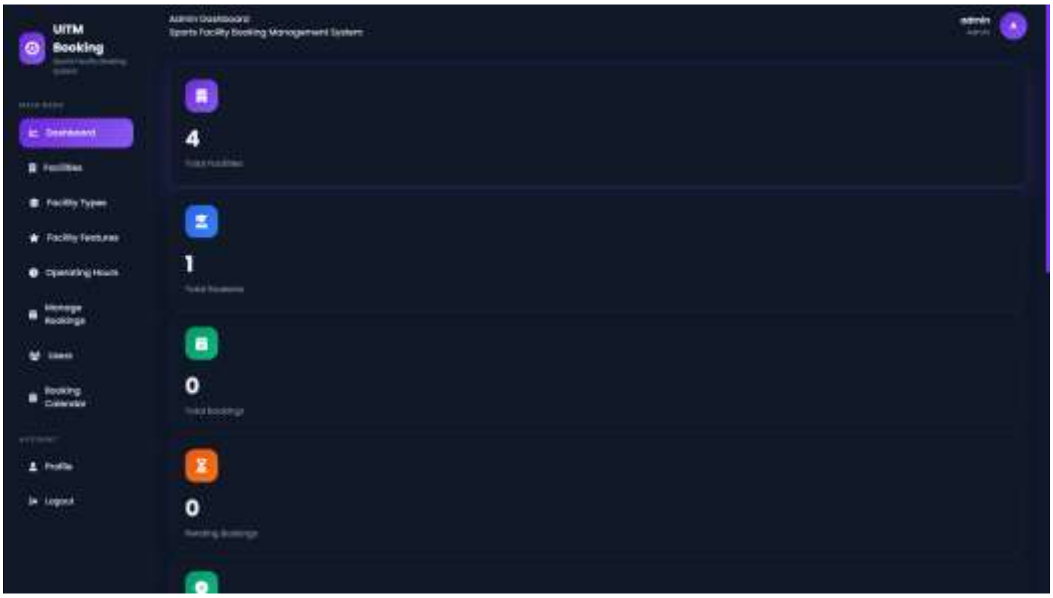

Figure 2.6 Responsive Dashboard Interface

2.8 Full Calendar

FullCalendar is a JavaScript library used to display events within an interactive calendar interface. It allows users to view schedules in monthly, weekly, or daily formats while providing visual representations of booking activities.

In the proposed system, FullCalendar is integrated into the administrator module to display sports facility bookings. Different booking statuses are represented using different colors, allowing administrators to monitor reservations efficiently and identify scheduling conflicts quickly.

The calendar module improves booking visualization and provides users with an organized overview of facility availability.

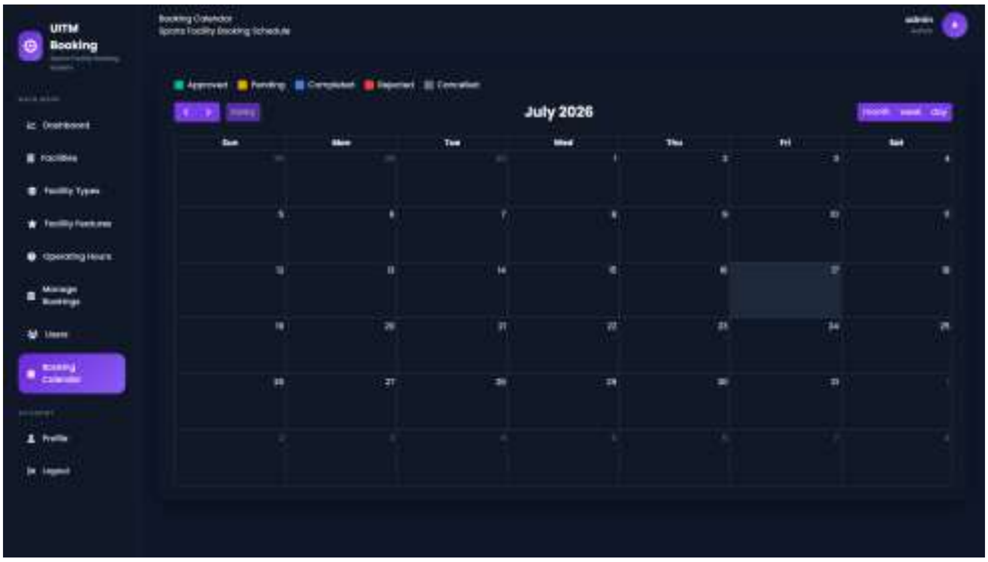

Figure 2.7 Booking Calendar Module

2.9 Authentication and Authorization

Authentication is the process of verifying user identity before granting access to the system, while authorization determines the permissions assigned to different users after successful authentication.

The proposed system implements Laravel Breeze to provide secure login, registration, password encryption, and session management. Two user roles are defined within the system: Administrator and Student. Each role has different access permissions according to its responsibilities.

This role-based access control enhances system security by ensuring that only authorized users can perform administrative functions.

User Role	Main Responsibilities
Administrator	Manage facilities, users, bookings, calendar, facility types, facility features and operating hours
Student	Book facilities, view booking history, manage profile

Table 2.2 User Roles and Permissions

2.10 Related Studies

Previous studies have shown that web-based booking systems improve operational efficiency by reducing manual processes and providing centralized information management. Researchers have reported that online reservation platforms help minimize booking conflicts, improve record management, and increase user satisfaction through real-time information access.

Most modern booking systems implement responsive web technologies, relational databases, and role-based authentication to improve security and usability. These findings support the development of the proposed Sports Facility Booking Management System using Laravel Framework.

2.11 Comparison of Existing Systems

Features	Google Calendar	CourtReserve	Proposed System
Online Booking	✓	✓	✓
Booking Approval	✗	✓	✓
Facility Management	✗	✓	✓
Dashboard Analytics	✗	Limited	✓
Calendar View	✓	✓	✓
Student Dashboard	✗	✓	✓
Laravel Framework	✗	✗	✓

Table 2.3 Comparison of Existing Booking Systems

2.12 Chapter Summary

This chapter reviewed the concepts, technologies, and previous studies that support the development of the Sports Facility Booking Management System. Laravel Framework, Bootstrap, MySQL, FullCalendar, and Chart.js were selected because they provide the necessary features for developing a secure, responsive, and efficient web application. The review of existing systems also demonstrates that the proposed system offers a comprehensive solution by integrating facility management, booking approval, booking conflict validation, and calendar scheduling into a single platform. These findings provide a strong theoretical foundation for the methodology and system design presented in the following chapter.

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter discusses the analysis and design of the proposed Sports Facility Booking Management System. The purpose of this chapter is to explain the methodology, system requirements, system architecture, database structure, and overall system design before implementation. Proper planning and system design are essential to ensure that the developed application fulfills the objectives identified in Chapter One.

The Sports Facility Booking Management System was designed using the Laravel Framework, which adopts the Model-View-Controller (MVC) architecture. This architecture separates the application into three major components, allowing better code organization, maintainability, and scalability. MySQL was selected as the database management system to store user information, booking records, facility details, operating hours, and other related data securely.

This chapter also presents the functional and non-functional requirements, software development methodology, system architecture, use case diagrams, activity diagrams, sequence diagrams, entity relationship diagram (ERD), database design, and user interface design. These designs serve as the blueprint for developing the final web application.

3.2 System Requirement Analysis

System requirement analysis identifies the functionalities and constraints required by the proposed Sports Facility Booking Management System. These requirements ensure that the developed system satisfies both administrator and student needs while providing an efficient and secure booking platform.

The requirements are divided into two categories:

- Functional Requirements
- Non-Functional Requirements

3.2.1 Functional Requirements

Functional requirements describe the operations and services that the system must perform.

Administrator Functional Requirements

The administrator is responsible for managing the entire booking system.

The administrator shall be able to:

- Login securely into the system.
- Manage student and administrator accounts.
- Add, edit and delete sports facilities.
- Manage facility types.
- Manage facility features.
- Manage operating hours.

- View all booking requests.
- Approve booking requests.
- Reject booking requests with remarks.
- Mark approved bookings as completed.
- View booking calendar.
- View dashboard statistics.
- Update administrator profile.
- Change password.
- Logout securely.

Student Functional Requirements

Students are the primary users of the reservation system.

Students shall be able to:

- Register an account.
- Login securely.
- View personal dashboard.
- Browse available sports facilities.
- Submit booking requests.
- View booking history.
- Cancel pending bookings.
- View booking calendar.
- Update profile information.
- Change password.
- Logout securely.

Booking Functional Requirements

The booking module shall:

- Validate booking information.
- Prevent overlapping bookings for the same facility.
- Store booking history.
- Record booking status.
- Allow administrator approval or rejection.

- Record administrator remarks for rejected bookings.
- Allow approved bookings to be marked as completed.

Module	Description
Authentication	Login, Registration, Logout
Dashboard	Display booking statistics
Facilities	Manage sports facilities
Facility Types	Manage facility categories
Facility Features	Manage facility equipment
Operating Hours	Configure facility schedules
Users	Manage administrator and student accounts
Booking	Submit, approve, reject and complete bookings
Calendar	Display facility reservation schedules
Profile	Update user profile and password

Table 3.1 Functional Requirements

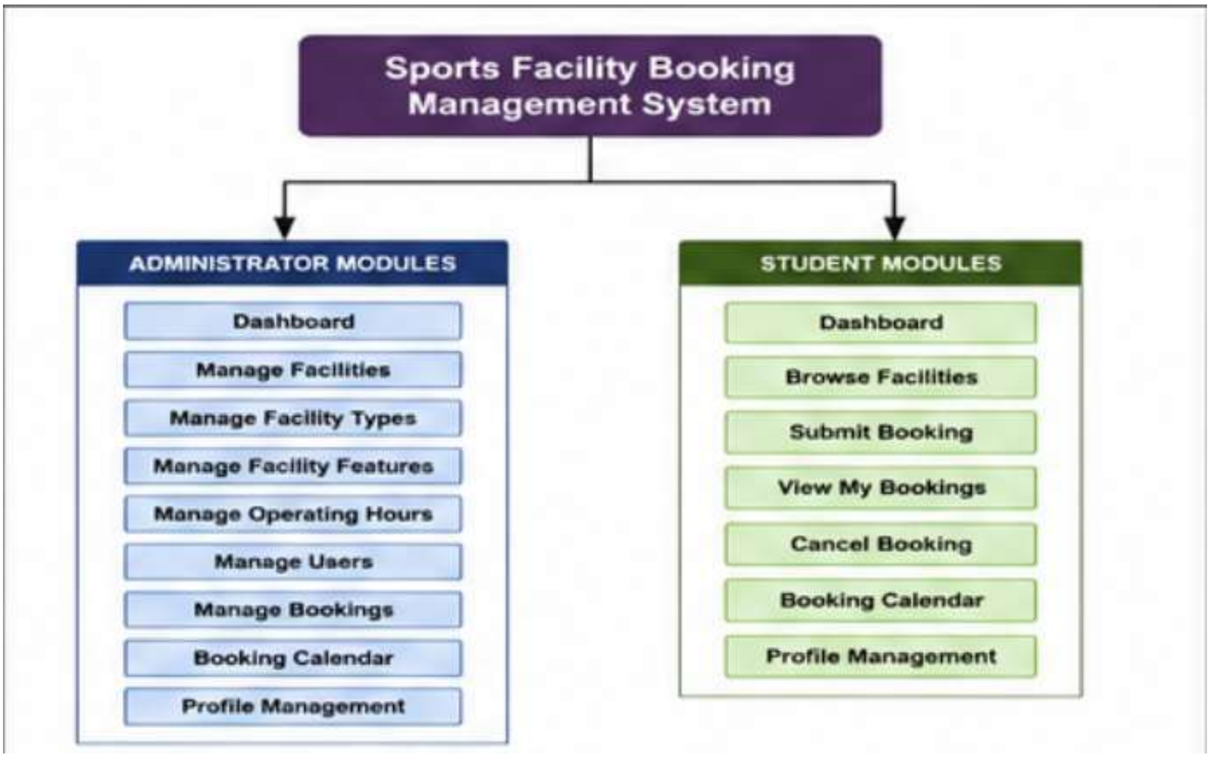

Figure 3.1 Functional Modules of the Proposed System

3.2.2 Non-Functional Requirements

Besides functional requirements, the proposed system must also satisfy several non-functional requirements.

Usability

The system shall provide a simple, responsive and user-friendly interface that allows users to navigate the application without extensive training.

Reliability

The system shall store booking records accurately while preventing duplicate and overlapping reservations.

Security

The system shall provide:

- Authentication
- Password encryption
- Role-based authorization
- Session management
- CSRF protection

using Laravel Framework.

Performance

The system shall respond quickly when:

- submitting bookings,
- retrieving booking history,
- loading dashboard information,
- searching booking records.

Maintainability

The MVC architecture allows developers to modify system components independently without affecting the entire application.

Scalability

The system can easily accommodate additional facilities, users and future modules without major modification.

Requirement	**Description**
Usability	Easy to use interface
Reliability	Accurate booking records
Security	Authentication and authorization
Performance	Fast response time
Maintainability	MVC architecture

Scalability	Future expansion supported

Table 3.2 Non-Functional Requirements

3.3 Software Development Methodology

The Sports Facility Booking Management System was developed using the **Waterfall Software Development Life Cycle (SDLC)** model.

The Waterfall model was selected because the project requirements were clearly defined before development began. Each development phase was completed sequentially, ensuring systematic project management and minimizing development risks.

The Waterfall model consists of five major phases:

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Maintenance

Requirement Analysis

The first phase involved identifying the problems associated with manual sports facility booking. User requirements were collected and analysed to determine the functions required by both administrators and students.

System Design

The second phase involved designing:

- database structure
- user interfaces
- system architecture
- use case diagrams
- activity diagrams
- sequence diagrams
- entity relationship diagram

Implementation

The implementation phase involved coding the proposed system using:

- Laravel Framework
- PHP
- MySQL
- Bootstrap
- JavaScript
- FullCalendar

Testing

Testing was conducted to verify that every module operated according to the specified requirements.

Functional testing included:

- User Login
- Registration
- Booking Submission
- Booking Conflict Detection
- Booking Approval
- Booking Rejection
- Booking Completion
- CRUD Operations

Maintenance

Future maintenance activities include:

- bug fixing
- performance improvement
- security updates
- future feature enhancements

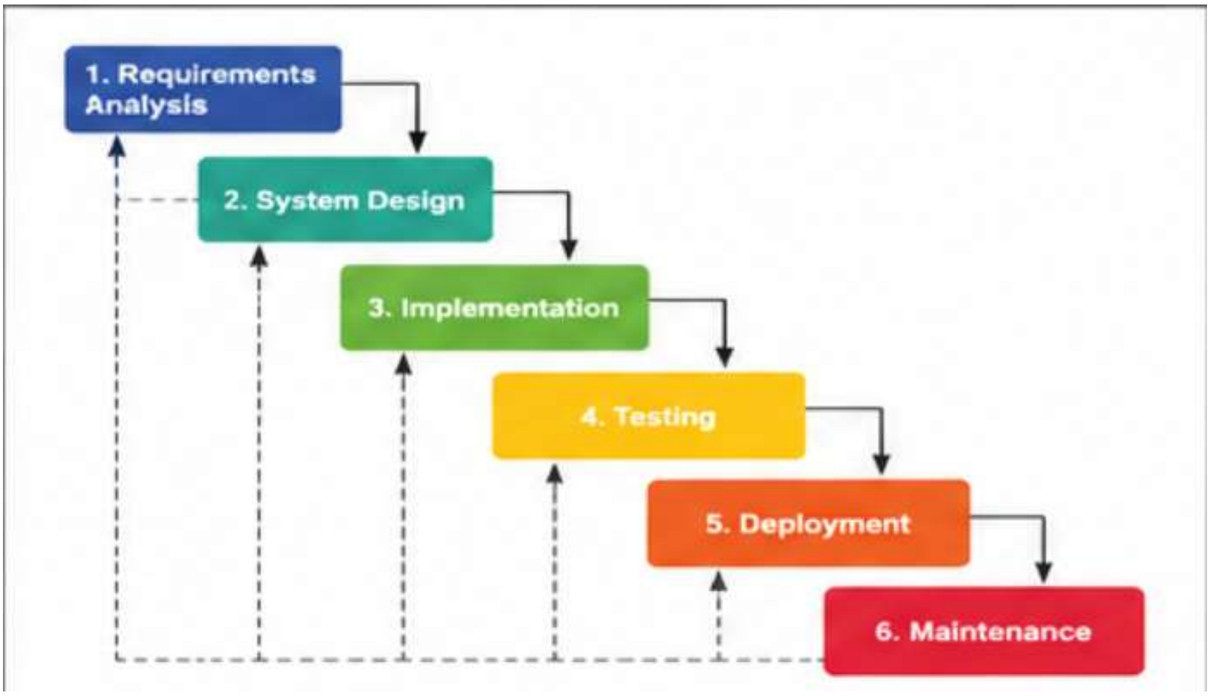

Figure 3.2 Waterfall Model

3.4 System Architecture

The proposed system adopts the **Model-View-Controller (MVC)** architecture provided by Laravel Framework.

MVC separates the application into three components.

Model

The Model communicates directly with the MySQL database.

Examples include:

- User
- Booking
- Facility
- Facility Type
- Facility Feature
- Operating Hour

View

The View consists of Blade templates that present information to administrators and students.

Examples include:

- Dashboard
- Login
- Booking Form
- Facility Management
- Calendar
- Profile

Controller

Controllers receive user requests, process business logic and communicate with the database.

Examples include:

- BookingController
- FacilityController
- UserController
- ProfileController

Advantages of MVC

- Better code organization.

- Easier maintenance.
- Reusable components.
- Improved scalability.
- Separation of concerns.

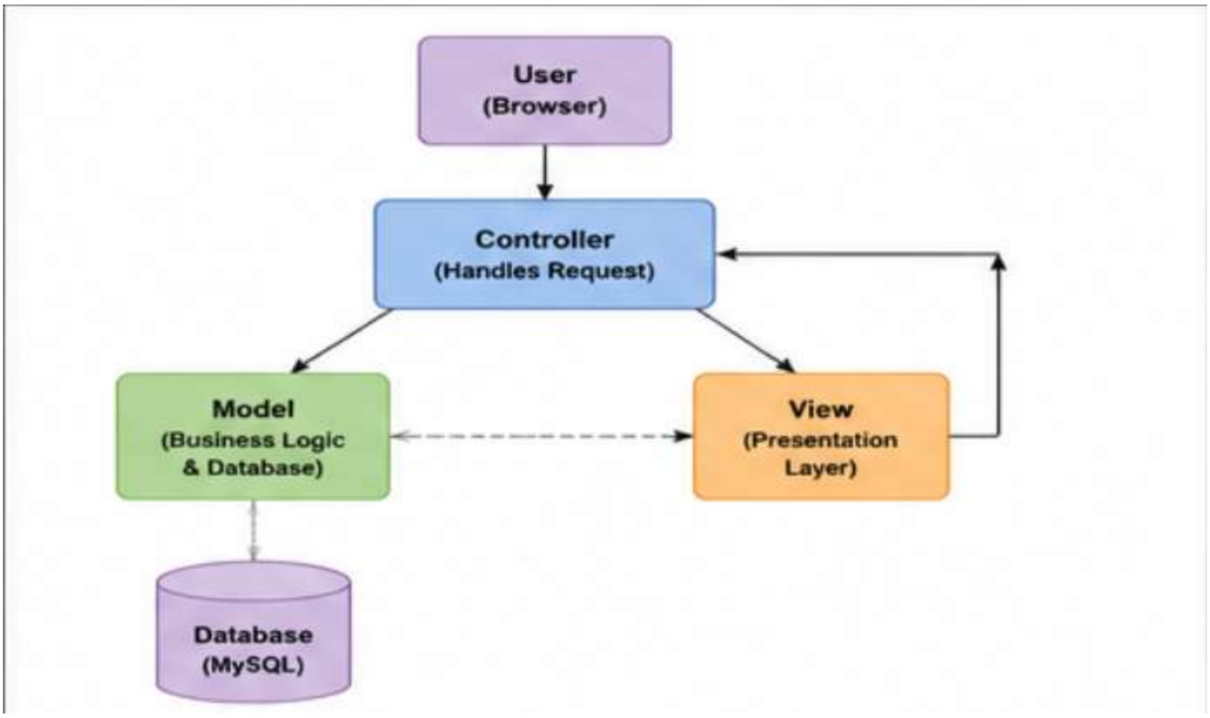

Figure 3.3 MVC Architecture

3.5 Use Case Diagram

The Sports Facility Booking Management System consists of two user roles:

- Administrator
- Student

Each role performs different operations according to the permissions assigned.

The following sections illustrate the interactions between users and the proposed system through use case diagrams.

Use Case Diagrams illustrate the interactions between users and the Sports Facility Booking Management System. The proposed system consists of two user roles: **Administrator** and **Student**. Each user performs different functions according to the access permissions assigned by the system.

The diagrams below summarize the major functions available to each user role.

3.5.1 Administrator Use Case Diagram

The Administrator has full control over the system. Besides managing sports facilities and users, the administrator is responsible for processing booking requests and maintaining overall system operations.

The administrator is able to perform the following functions:

- Login
- Manage Facilities
- Manage Facility Types
- Manage Facility Features
- Manage Operating Hours
- Manage Users
- View Dashboard
- Approve Booking
- Reject Booking
- Complete Booking
- View Booking Calendar
- Update Profile
- Logout

The Administrator Use Case Diagram is shown in **Figure 3.4**.

Figure 3.4 Administrator Use Case Diagram Screenshot Required

3.5.2 Student Use Case Diagram

Students interact with the system to reserve sports facilities and manage their own bookings.

The Student can:

- Register

- Login
- View Dashboard
- Browse Facilities
- Submit Booking
- View Booking History
- Cancel Booking
- View Booking Calendar
- Update Profile
- Logout

The Student Use Case Diagram is shown in **Figure 3.5**.

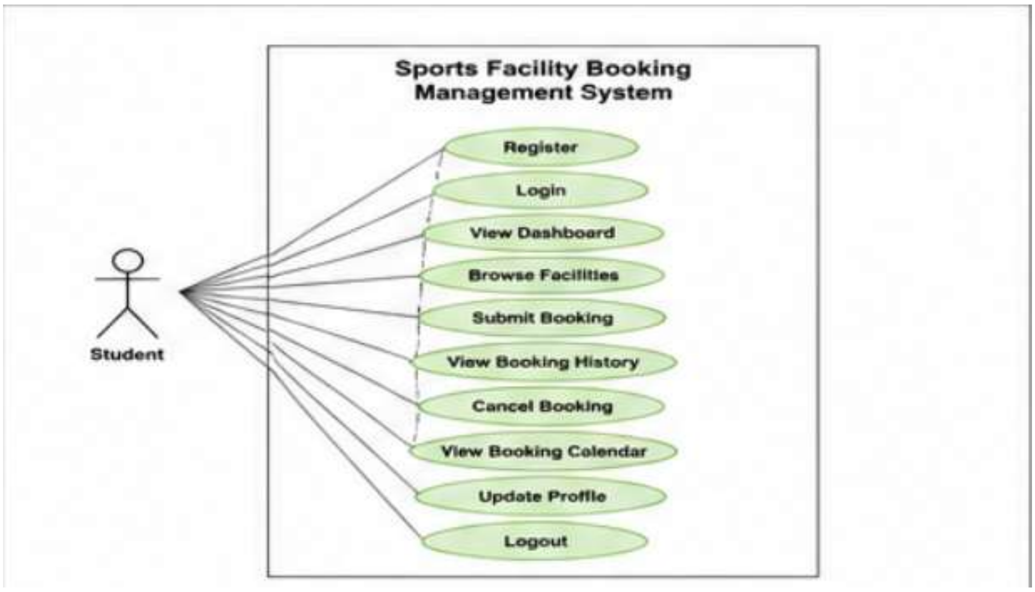

Figure 3.5 Student Use Case Diagram

3.6 Activity Diagram

Activity Diagrams describe the workflow performed by users when interacting with the proposed system.

3.6.1 Student Booking Activity Diagram

The booking process begins when a student logs into the system and selects a sports facility. The student enters the booking details, including the booking date, start time, end time, and booking purpose.

The system validates all booking information. It checks whether the selected facility is already booked during the requested time slot. If a booking conflict exists, the system displays an error message and prevents the booking from being submitted. Otherwise, the booking is saved with a **Pending** status.

The booking request is then forwarded to the administrator for approval.

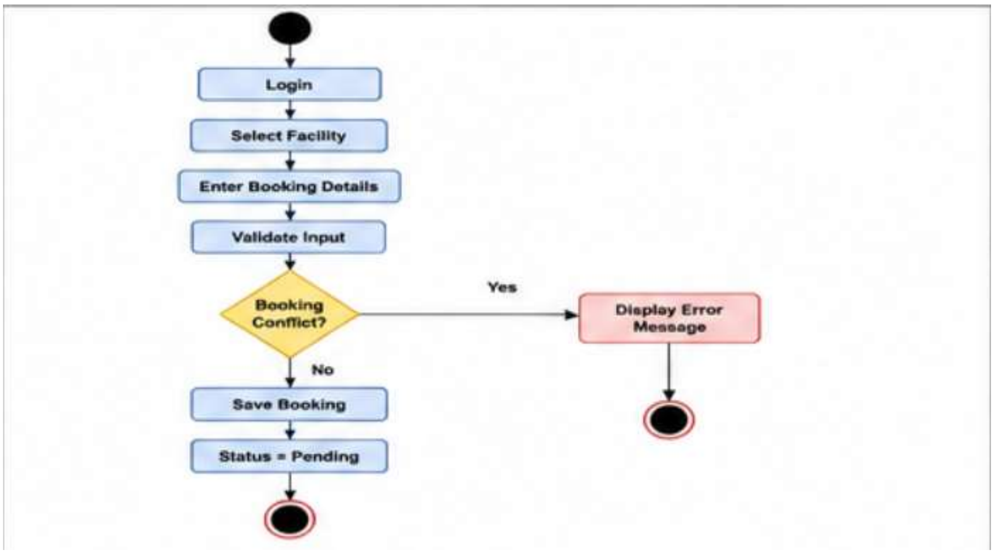

Figure 3.6 Student Booking Activity Diagram

3.6.2 Administrator Booking Activity Diagram

After receiving a booking request, the administrator reviews the booking details.

The administrator may:

- Approve Booking
- Reject Booking
- Complete Booking (after facility usage)

If rejected, remarks must be entered before the booking status changes to **Rejected**.

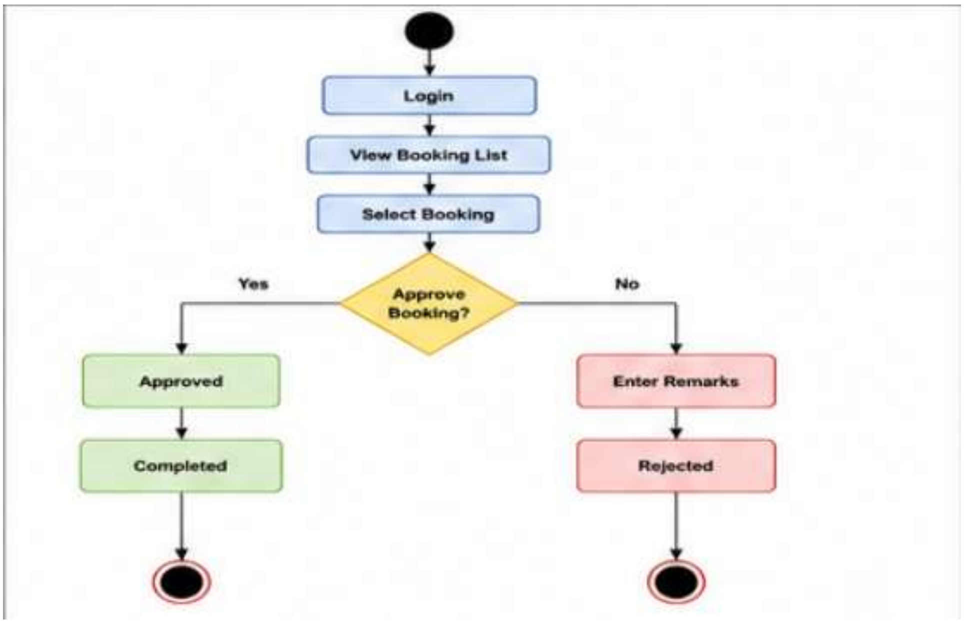

Figure 3.7 Administrator Booking Activity Diagram

3.7 Sequence Diagram

Sequence diagrams describe the interaction between users and the proposed system over time.

3.7.1 Booking Submission Sequence Diagram

The booking submission process involves:

- Student
- Web Browser
- Booking Controller
- MySQL Database

The sequence begins when the student submits a booking request. The controller validates the booking information and checks for overlapping bookings. If validation succeeds, the booking record is stored in the database and a confirmation message is displayed.

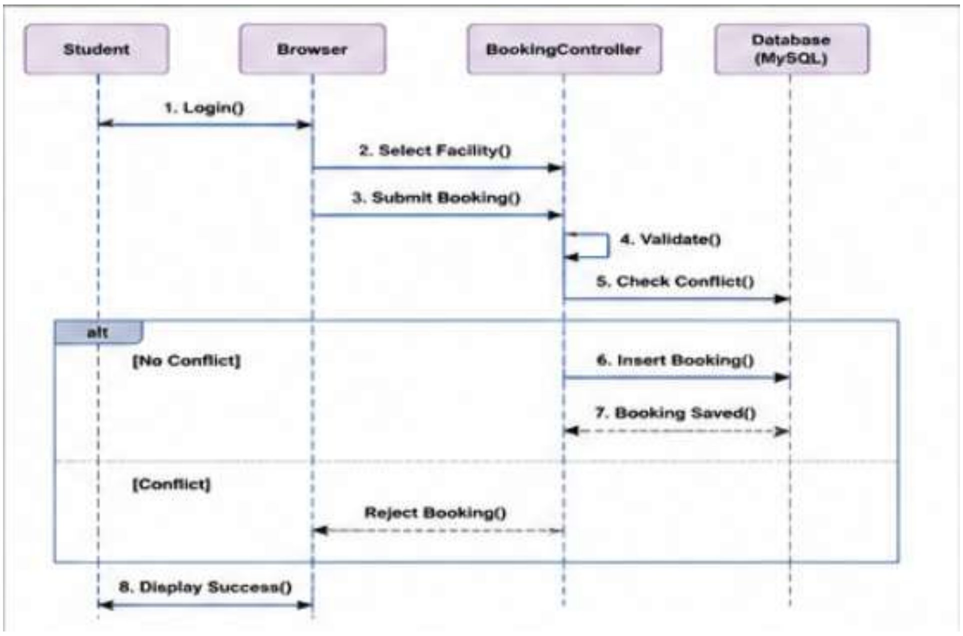

Figure 3.8 Booking Submission Sequence Diagram

3.7.2 Booking Approval Sequence Diagram

The booking approval sequence begins when the administrator reviews pending bookings.

The administrator selects a booking request and either approves or rejects it. If rejected, remarks are recorded before updating the booking status.

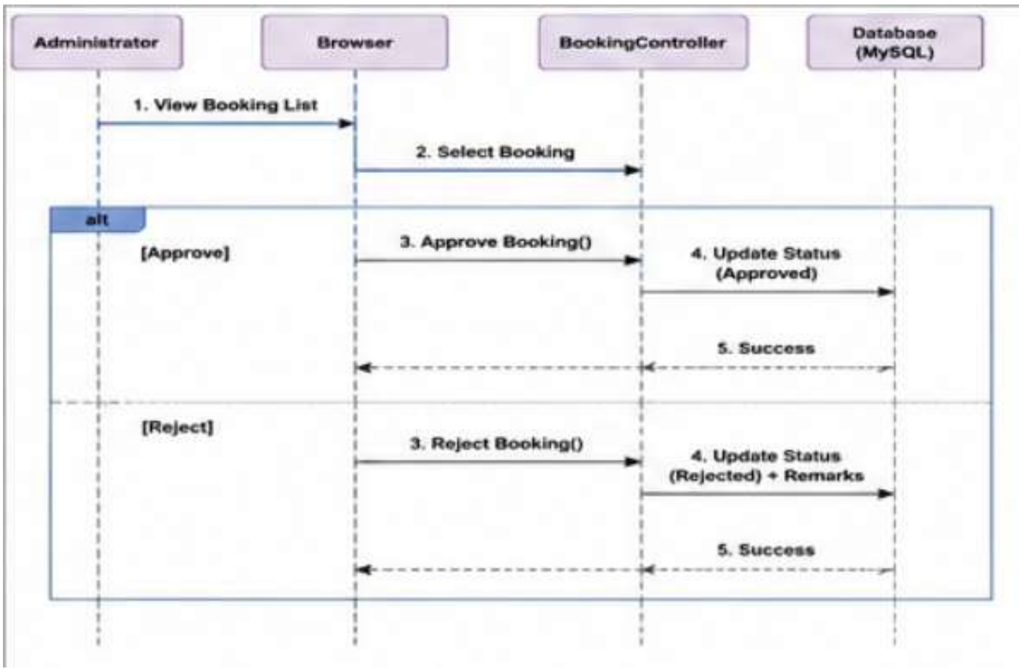

Figure 3.9 Booking Approval Sequence Diagram

3.8 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) illustrates the relationship among database tables used in the Sports Facility Booking Management System.

The proposed system consists of six primary entities:

- Users
- Facilities
- Facility Types
- Facility Features
- Operating Hours
- Bookings

Each booking is associated with one student and one sports facility through foreign key relationships. Facilities are further categorized using facility types and facility features, while operating hours define the available booking schedule for each facility.

The ERD ensures data integrity by minimizing redundancy and establishing relationships between related entities.

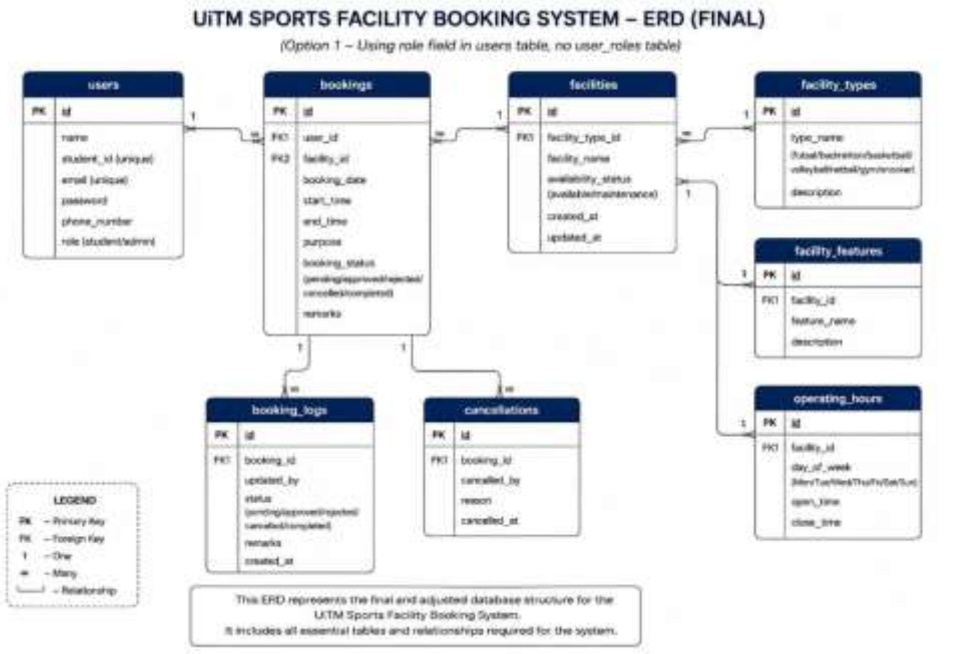

Figure 3.10 Final Entity Relationship Diagram

3.9 Database Design

The Sports Facility Booking Management System uses MySQL as the database management system. The database consists of several interconnected tables that store information related to users, sports facilities, bookings, facility types, facility features, and operating hours.

Field	**Description**
id	Primary Key
name	User Name
student_id	Student / Staff ID
email	Email Address
phone_number	Contact Number
role	Administrator / Student
password	Encrypted Password

Table 3.3 Users Table

Field	**Description**
id	Primary Key
facility_name	Sports Facility
facility_type_id	Facility Category
availability_status	Availability
description	Description

Table 3.4 Facilities Table

Field	Description
id	Primary Key
type_name	Facility Type

Table 3.5 Facility Types Table

Field	Description
id	Primary Key
feature_name	Facility Feature

Table 3.6 Facility Features Table

Field	Description
id	Primary Key
facility_id	Facility
opening_time	Opening Time
closing_time	Closing Time

Table 3.7 Operating Hours Table

Field	Description
id	Primary Key
user_id	Student
facility_id	Facility
booking_date	Booking Date
start_time	Start Time
end_time	End Time
purpose	Booking Purpose
booking_status	Pending / Approved / Rejected / Completed / Cancelled
remarks	Administrator Remarks

Table 3.8 Bookings Table

Table	Action						Rows	Type	Collation	Size	Overhead
bookings	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
booking_logs	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
cache	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
cache_locks	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
cancellations	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
facilities	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
facility_features	Browse	Structure	Search	Insert	Empty	Drop	3	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
facility_types	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
failed_jobs	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
jobs	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
job_batches	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
migrations	Browse	Structure	Search	Insert	Empty	Drop	10	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
operating_hours	Browse	Structure	Search	Insert	Empty	Drop	3	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
password_reset_tokens	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
sessions	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
users	Browse	Structure	Search	Insert	Empty	Drop	3	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
16 tables	Sum						31	InnoDB	utf8mb4_general_ci	528.0 KiB	0 B

Figure 3.11 Database Structure

3.10 User Interface Design

The user interface was designed using Bootstrap 5 to provide a responsive and user-friendly experience across desktop and mobile devices. The interface follows a consistent dark-themed layout with purple accents to improve visual appearance and usability.

The main interfaces developed include:

- Login Page
- Registration Page
- Administrator Dashboard
- Student Dashboard
- Facility Management
- User Management
- Booking Management
- Booking Calendar
- Profile Management

Each interface was developed to ensure users can perform tasks efficiently while maintaining consistency throughout the system.

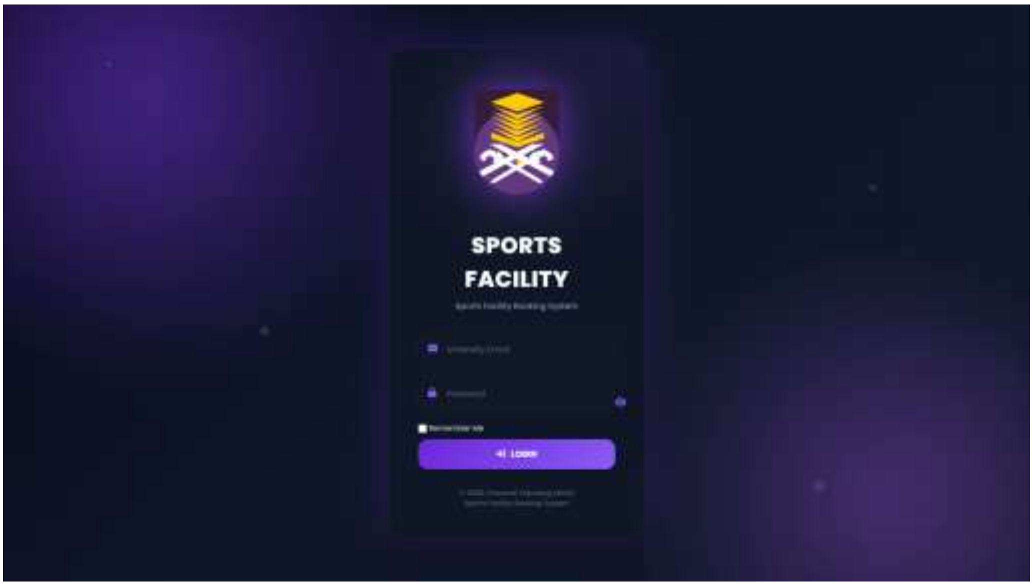

Figure 3.12 Login Page

Figure 3.13 Administrator DashboardFigure

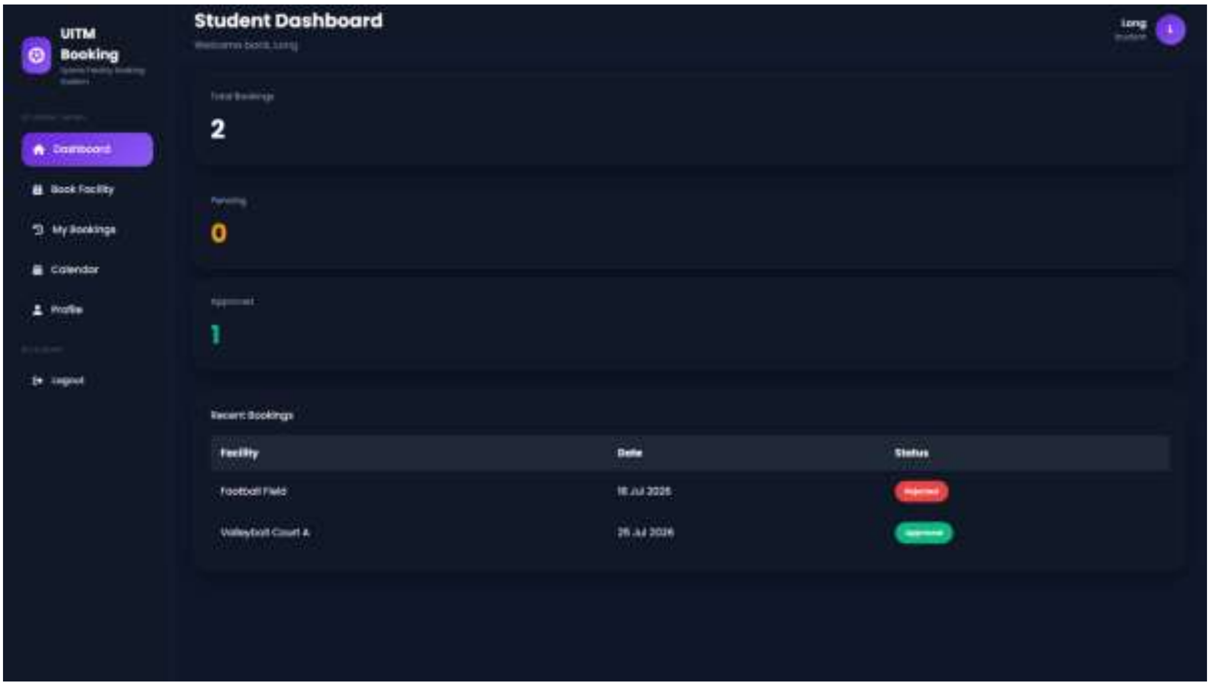

Figure 3.14 Student DashboardFigure

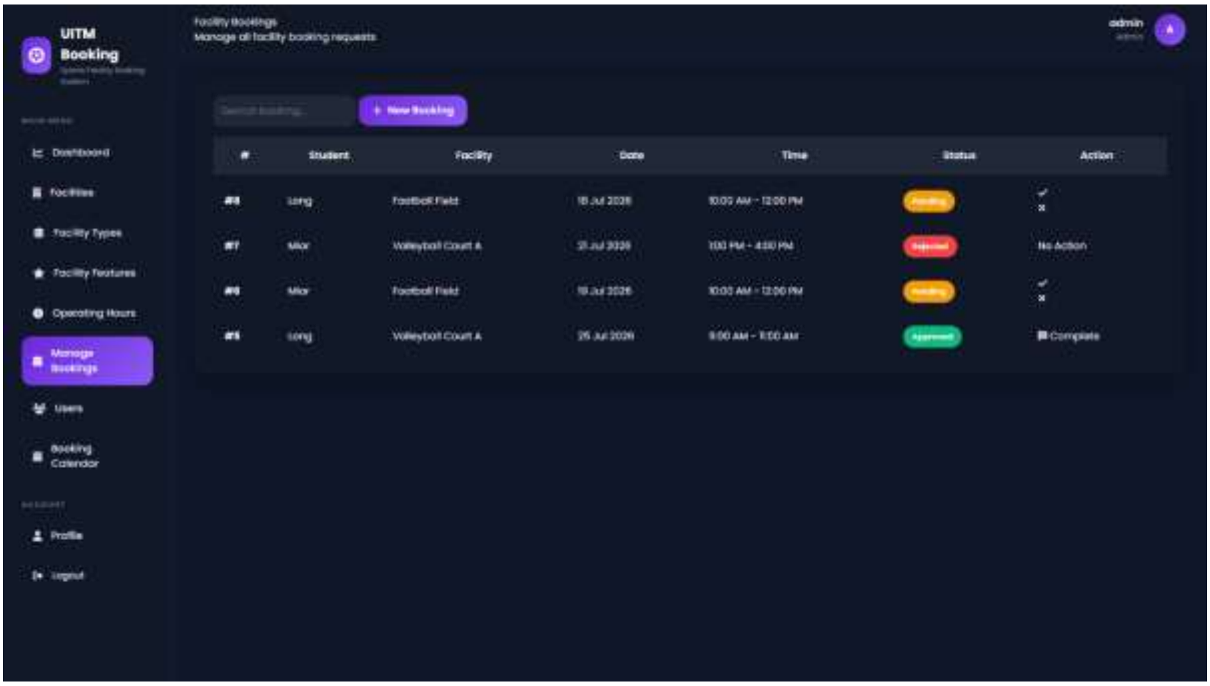

Figure 3.15 Booking Management

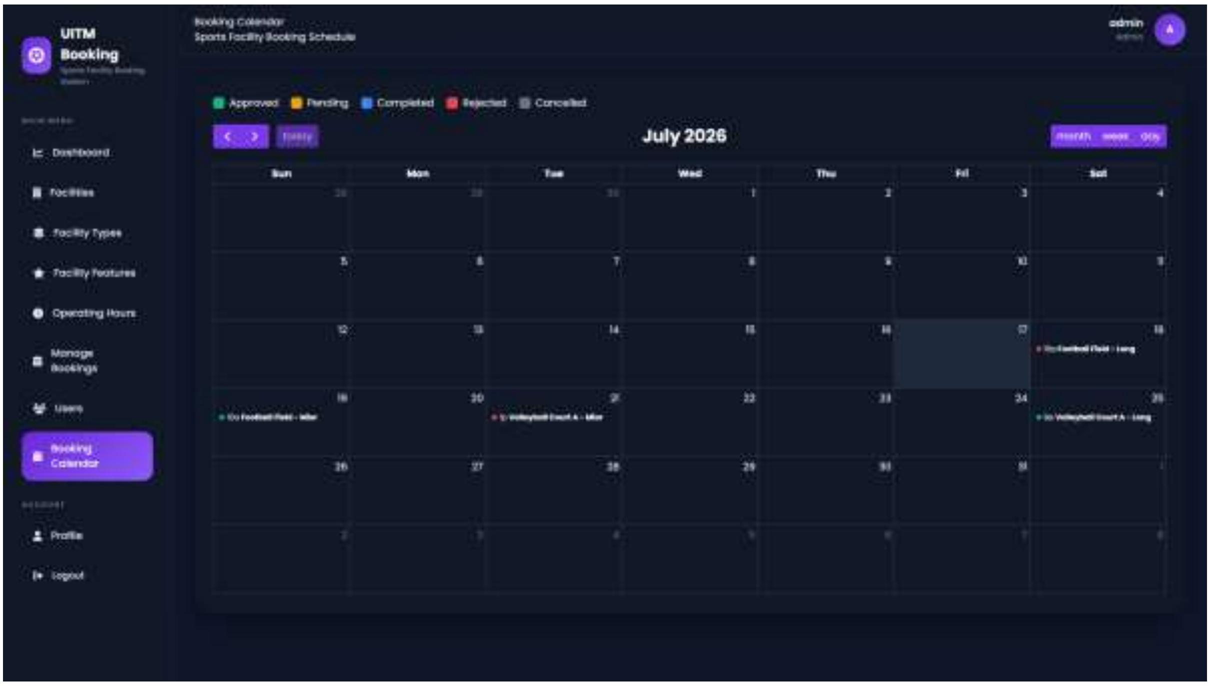

Figure 3.16 Booking Calendar

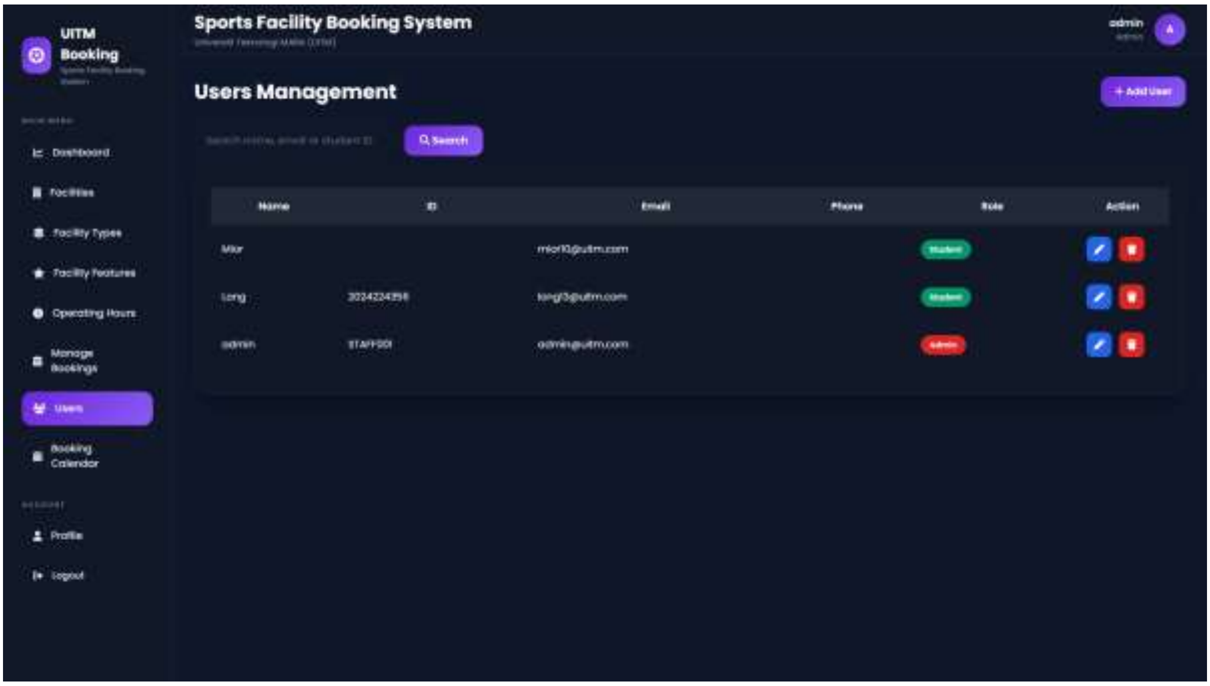

Figure 3.17 User Management

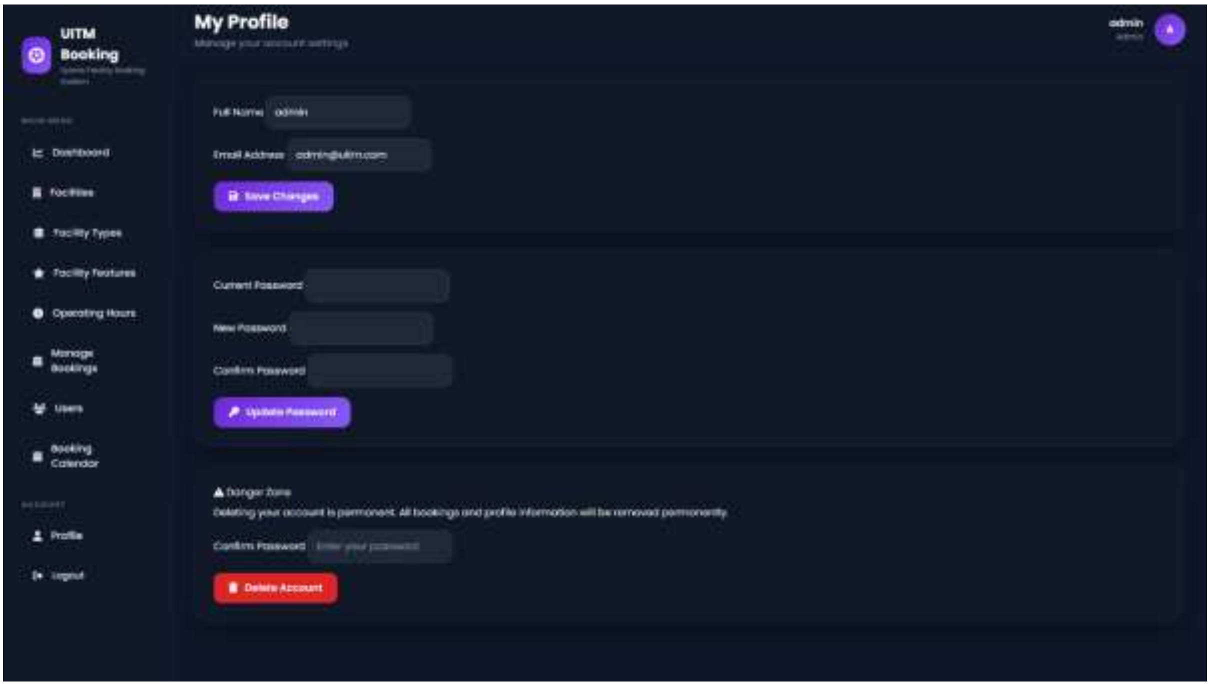

Figure 3.18 Profile Management

3.11 Chapter Summary

This chapter discussed the analysis and design of the proposed Sports Facility Booking Management System. The system requirements, software development methodology, system architecture, use case diagrams, activity diagrams, sequence diagrams, entity relationship diagram, database design, and user interface design were presented to provide a comprehensive blueprint for system development. The use of the Laravel Framework following the Model-View-Controller (MVC) architecture contributes to a structured, maintainable, and scalable web application. The designs and analyses described in this chapter provide a strong foundation for the implementation phase presented in the next chapter.

SYSTEM IMPLEMENTATION

4.1 Introduction

This chapter discusses the implementation of the Sports Facility Booking Management System developed using the Laravel Framework. It explains how the system was developed based on the analysis and design presented in Chapter Three. The implementation process involved translating the system requirements into a fully functional web-based application using Laravel, MySQL, Bootstrap, JavaScript, and FullCalendar.

The system consists of two user roles, namely **Administrator** and **Student**, each with different access permissions. Various modules were implemented to manage sports facilities, facility types, facility features, operating hours, user accounts, booking activities, and personal profiles. In addition, booking conflict validation was implemented to prevent overlapping reservations for the same facility during the same time period.

This chapter also presents screenshots of the developed interfaces together with explanations of each module to demonstrate how the system fulfills the objectives of the project.

4.2 Development Environment

The Sports Facility Booking Management System was developed using several software tools and technologies. These tools provide a stable environment for designing, developing, testing, and deploying the application.

Component	Software
Operating System	Windows 11
IDE	Visual Studio Code
Programming Language	PHP 8.2
Framework	Laravel 12
Database	MySQL
Web Server	XAMPP (Apache)
Front-end	Bootstrap 5
JavaScript Library	FullCalendar
Version Control	Git & GitHub
Browser	Google Chrome

Table 4.1 Development Environment

The combination of these technologies provides a modern web application that is secure, responsive, and easy to maintain.

4.3 System Modules

The Sports Facility Booking Management System consists of several integrated modules that work together to support the online reservation process.

4.3.1 Login Module

The Login Module provides secure authentication for both administrators and students. Users are required to enter their registered email address and password before accessing the system.

Laravel Breeze is used to implement secure authentication together with encrypted passwords and session management.

Successful authentication redirects users according to their assigned role.

Administrator

↓

Administrator Dashboard

Student

↓

Student Dashboard

Figure 4.1 Login Interface

4.3.2 Student Registration Module

The registration module allows new students to create an account by providing their personal information.

The required information includes:

- Full Name
- Student ID
- Email Address

- Phone Number
- Password

The system validates all information before storing it in the database.

Duplicate email addresses are not permitted.

Figure 4.2 Student Registration Page

4.3.3 Administrator Dashboard

After successful login, administrators are redirected to the dashboard.

The dashboard summarizes important system information such as:

- Total Facilities
- Total Students
- Total Bookings
- Pending Bookings
- Approved Bookings
- Completed Bookings

Recent booking requests are also displayed to allow administrators to monitor system activities efficiently.

Figure 4.3 Administrator Dashboard

4.3.4 Student Dashboard

Students are presented with a personalized dashboard after login.

The dashboard displays:

- Total Bookings
- Pending Bookings
- Approved Bookings
- Recent Booking History

This allows students to monitor their reservation activities conveniently.

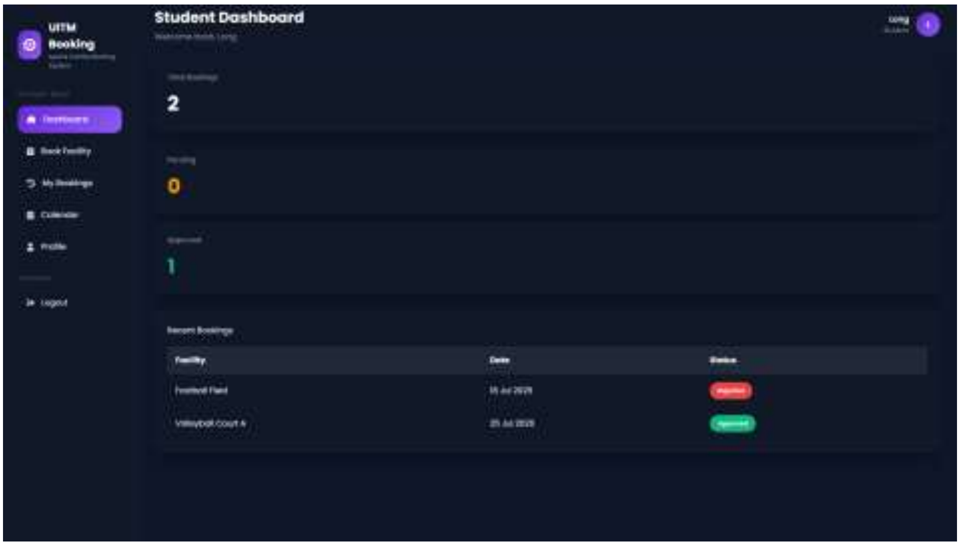

Figure 4.4 Student Dashboard

4.3.5 Facility Management Module

The Facility Management Module allows administrators to manage all sports facilities available in the system.

Administrators can:

- Add New Facility
- Edit Facility
- Delete Facility
- Search Facility

Each facility stores information such as:

- Facility Name
- Facility Type
- Availability Status
- Description

The module ensures that facility information remains accurate and up-to-date.

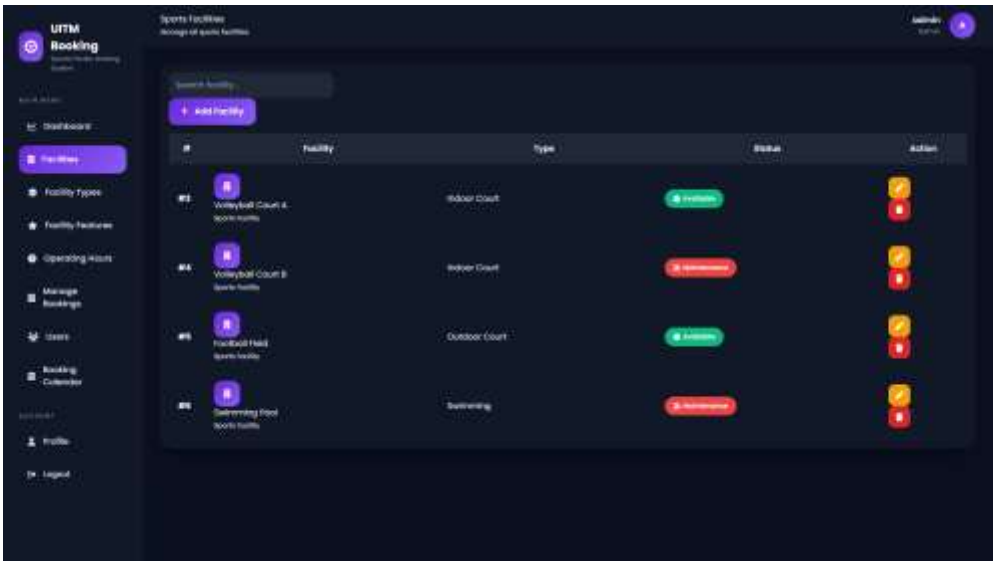

Figure 4.5 Facility Management

4.3.6 Facility Type Module

Facility Types are used to categorize sports facilities.

Examples include:

- Indoor Court
- Outdoor Court
- Hall
- Swimming Pool

Administrators may:

- Add Facility Type

- Edit Facility Type
- Delete Facility Type

This module improves data organization by grouping similar facilities.

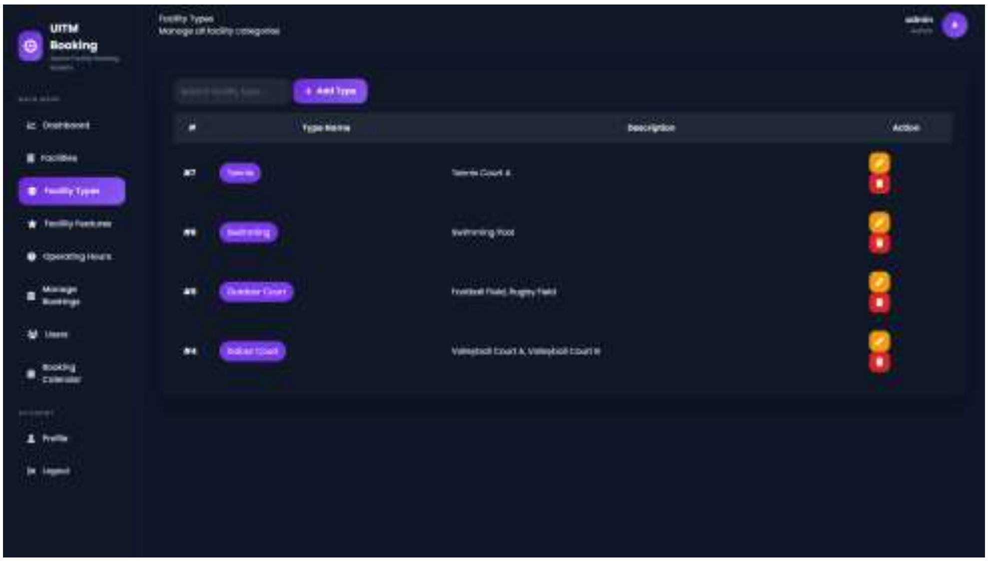

Figure 4.6 Facility Type Management

4.3.7 Facility Features Module

Facility Features describe the amenities available for each sports facility.

Examples include:

- Lighting
- Parking
- Changing Room
- Spectator Seating

Administrators can:

- Add Feature
- Edit Feature
- Delete Feature

This allows users to understand the available facilities before making reservations

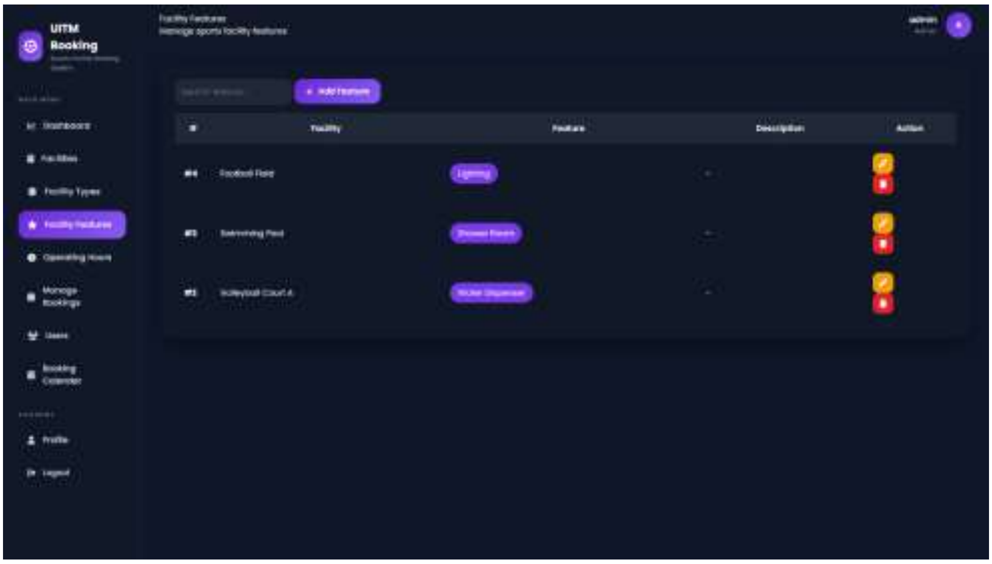

Figure 4.7 Facility Features

4.3.8 Operating Hours Module

Operating Hours define the available booking schedule for each sports facility.

Administrators may configure:

- Opening Time
- Closing Time

This ensures that bookings are made only within the operating schedule of the facility.

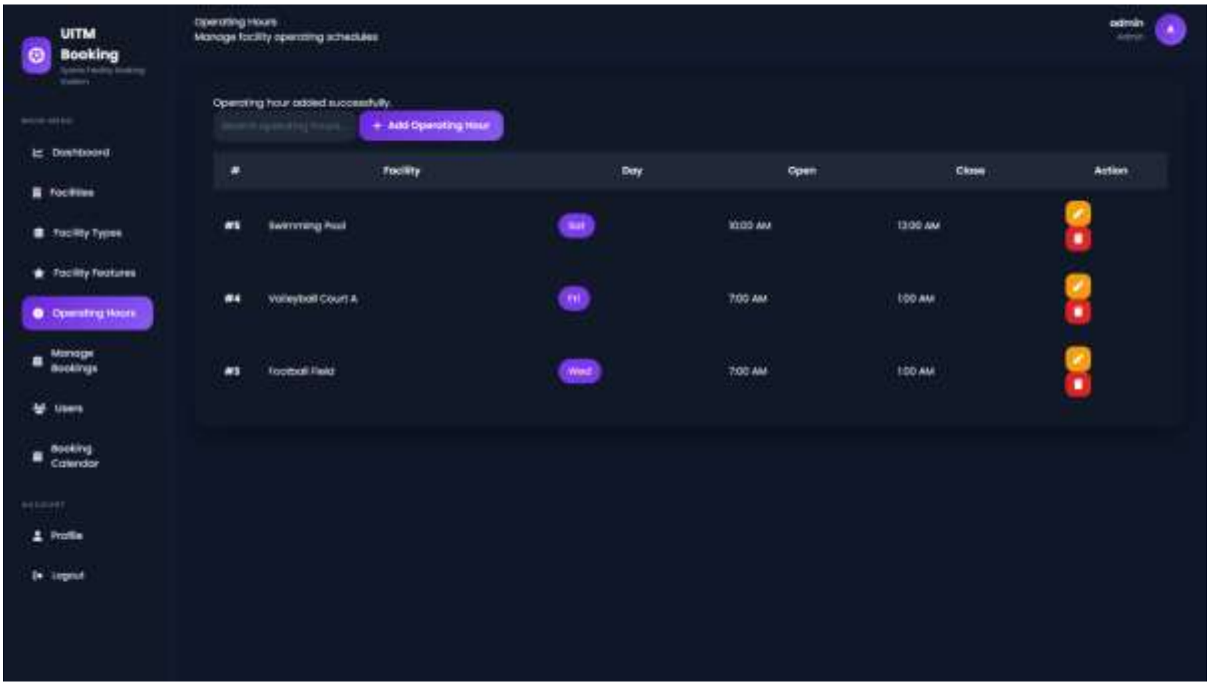

Figure 4.8 Operating Hours

4.3.9 User Management Module

The User Management Module allows administrators to manage all system users.

Administrators can:

- Add User
- Edit User
- Delete User
- Search User

The system supports two user roles:

- Administrator
- Student

The module also stores user information including:

- Name
- ID
- Email
- Phone Number
- Role

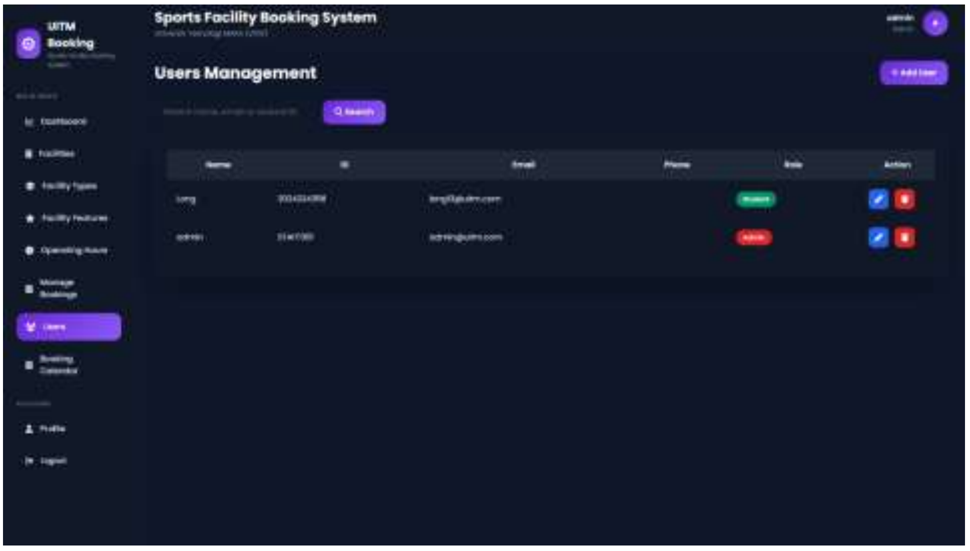

Figure 4.9 User Management

4.3.10 Booking Management Module

The Booking Management Module is the core component of the system.

Students submit booking requests, while administrators review every request.

The administrator may:

- Approve Booking
- Reject Booking
- Complete Booking

Rejected bookings require administrators to provide remarks before submission.

The booking status changes automatically according to the administrator's action.

Possible booking statuses include:

- Pending
- Approved
- Rejected
- Completed
- Cancelled

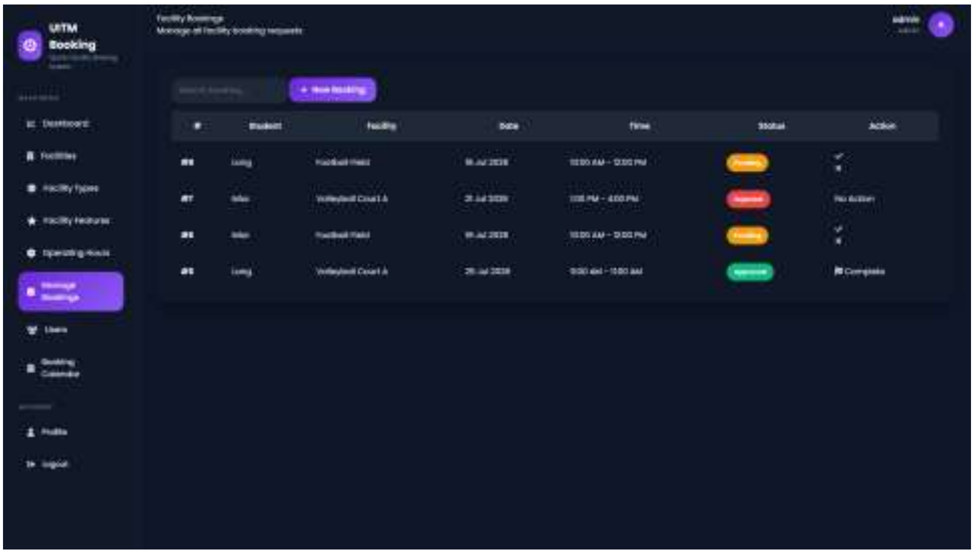

Figure 4.10 Booking Management

4.3.11 Booking Conflict Validation

One of the major improvements implemented in the system is automatic booking conflict validation.

Before storing a booking, the system checks whether another reservation already exists for:

- the same facility,
- the same booking date,
- overlapping booking times.

If a conflict is detected, the booking is rejected and an error message is displayed.

This feature prevents double-booking and improves booking accuracy.

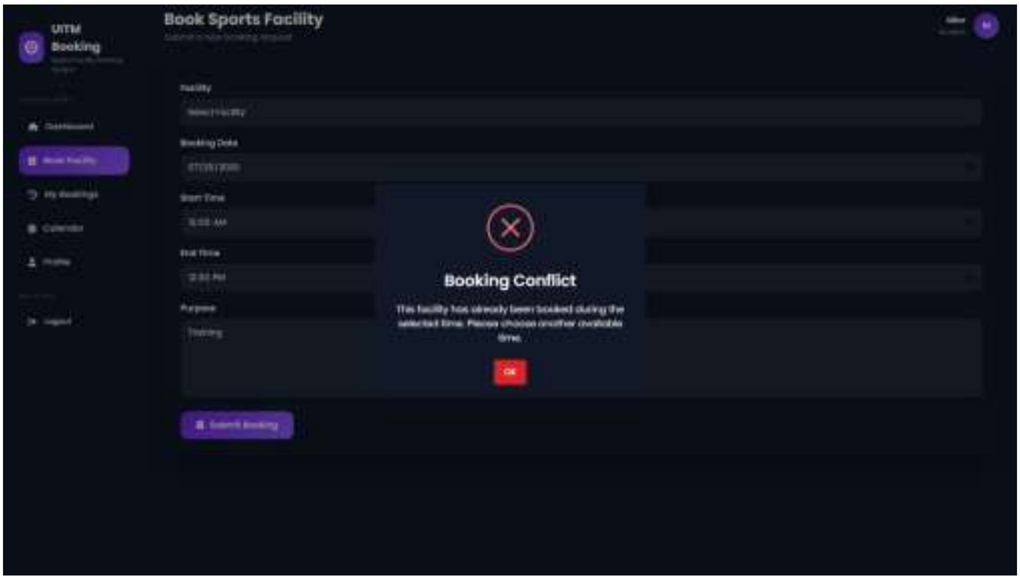

Figure 4.11 Booking Conflict Validation

4.3.12 Booking Calendar Module

The Booking Calendar Module provides administrators and students with a visual representation of facility reservations using the **FullCalendar** JavaScript library. Instead of viewing booking records in a table, users can view reservations based on calendar dates.

The calendar displays approved bookings according to the booking date and scheduled time, allowing users to identify reserved facilities more easily. Different booking entries are displayed within the monthly calendar view, providing a clear overview of facility usage.

This module assists administrators in monitoring reservation schedules while helping students check facility availability before submitting booking requests.

Figure 4.12 Booking Calendar

4.3.13 Profile Management Module

The Profile Management Module allows both administrators and students to update their personal information after logging into the system.

Users may modify:

- Full Name
- Staff ID / Student ID
- Email Address
- Phone Number
- Password

Laravel validates all submitted information before updating the database. Passwords remain encrypted using Laravel's built-in hashing mechanism to ensure account security.

The module also includes password change functionality and account deletion confirmation for additional account management.

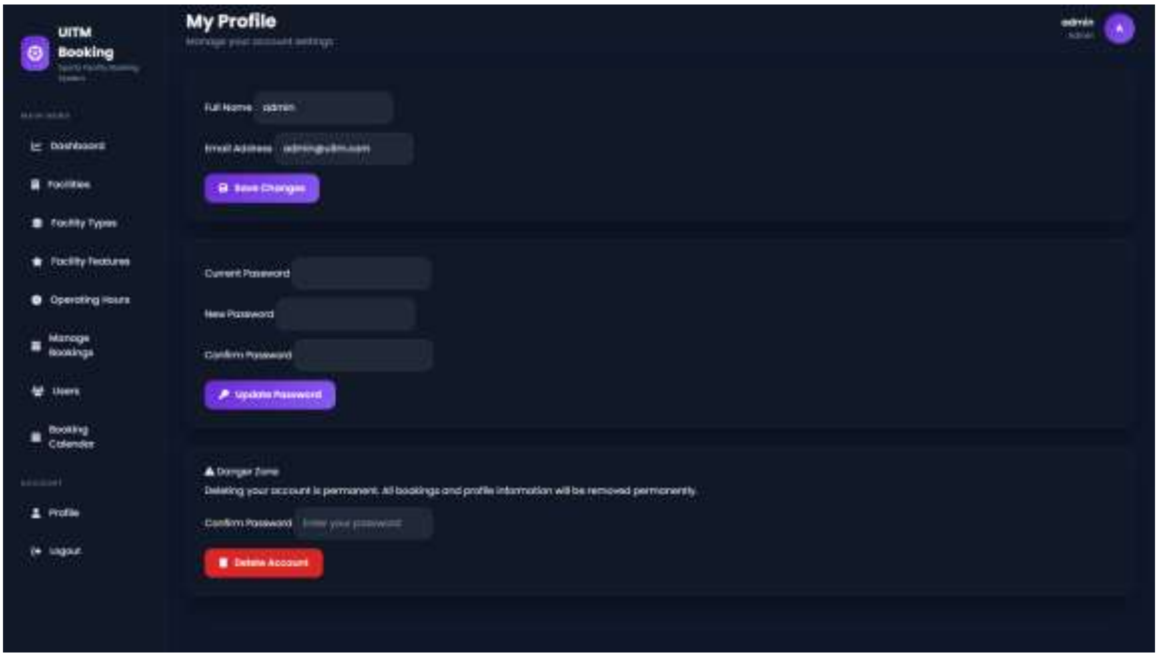

Figure 4.13 Profile Management

4.4 System Testing

System testing was conducted to verify that every module in the Sports Facility Booking Management System functions correctly according to the project requirements. Functional testing was performed by executing each feature and comparing the actual system behavior with the expected outcomes.

Test Case	**Expected Result**	**Actual Result**	**Status**
User Registration	Student account successfully created	Successfully created	Passed
User Login	User redirected to correct dashboard	Successful login	Passed
Facility Management	Administrator manages facilities	Successfully managed	Passed
Facility Type Management	Add/Edit/Delete facility types	Successfully managed	Passed
Facility Feature Management	Add/Edit/Delete facility features	Successfully managed	Passed
Operating Hours Management	Configure operating hours	Successfully updated	Passed
User Management	Add/Edit/Delete users	Successfully managed	Passed
Booking Submission	Booking successfully submitted	Successfully submitted	Passed
Booking Conflict Validation	Overlapping booking rejected	Conflict detected correctly	Passed
Booking Approval	Booking status changes to Approved	Successfully approved	Passed
Booking Rejection	Booking rejected with remarks	Successfully rejected	Passed

Booking Completion	Booking marked as Completed	Successfully completed	Passed
Booking Cancellation	Student cancels pending booking	Successfully cancelled	Passed
Booking Calendar	Booking displayed on calendar	Successfully displayed	Passed
Profile Update	User information updated	Successfully updated	Passed
Password Update	Password changed successfully	Successfully updated	Passed

Table 4.2 Functional Testing Results

4.4.1 Booking Conflict Testing

The booking conflict validation feature was tested by creating multiple booking requests for the same facility at overlapping times.

Test Scenario

Booking A

- Facility: Volleyball Court
- Date: 25 July 2026
- Time: 9:00 AM – 11:00 AM

Booking B

- Facility: Volleyball Court
- Date: 25 July 2026
- Time: 10:00 AM – 12:00 PM

Expected Result

The system rejects the second booking and displays a booking conflict message.

Actual Result

The booking request was successfully blocked, and a booking conflict warning was displayed using SweetAlert.

Status: Passed

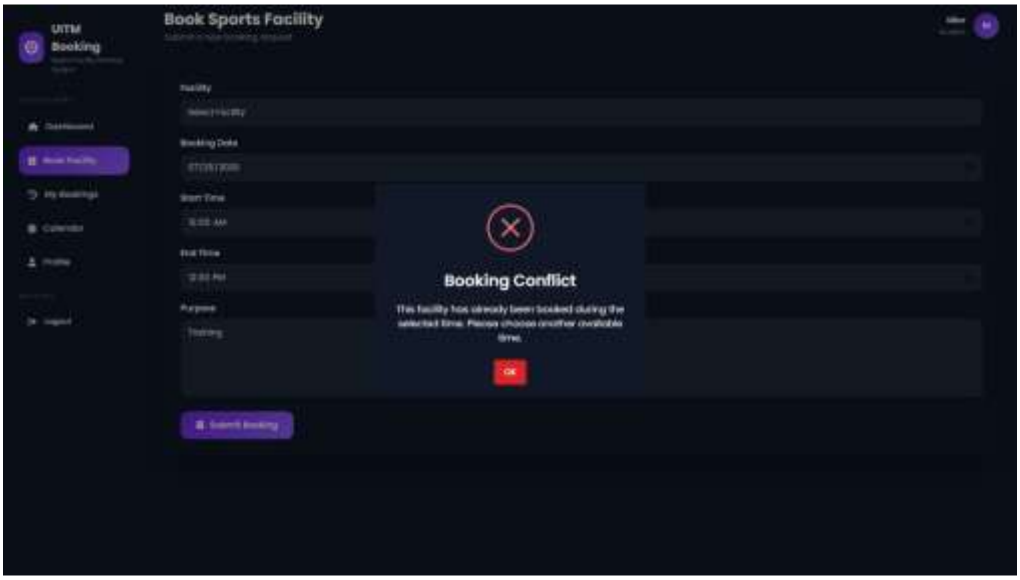

Figure 4.14 Booking Conflict Testing

4.4.2 User Acceptance Testing (UAT)

User Acceptance Testing (UAT) was conducted to evaluate the usability and functionality of the proposed system from the perspective of end users.

Both administrator and student roles successfully completed their assigned tasks without encountering critical errors. The testing confirmed that the developed system meets the functional requirements identified during the system analysis phase.

The overall testing results indicate that the Sports Facility Booking Management System operates as expected and is suitable for managing sports facility reservations within the intended environment.

4.5 Chapter Summary

This chapter presented the implementation of the Sports Facility Booking Management System. The implementation process covered the development of all major modules, including authentication, dashboard, facility management, user management, booking management, booking conflict validation, booking calendar, and profile management. Functional testing demonstrated that all implemented modules operated according to the specified requirements. The successful testing results indicate that the developed system is stable, reliable, and ready for practical use.

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter concludes the development of the Sports Facility Booking Management System. It summarizes the project achievements, discusses the strengths and limitations of the developed system, and presents several recommendations for future improvements.

The proposed system was successfully developed using the Laravel Framework following the Model-View-Controller (MVC) architecture. The implementation demonstrates how modern web technologies can improve sports facility reservation management within an educational institution.

5.2 Project Achievement

The objectives identified in Chapter One have been successfully achieved through the implementation of the proposed system.

Firstly, the system provides students with an online platform to submit sports facility booking requests conveniently without relying on manual procedures.

Secondly, administrators are provided with comprehensive management modules to manage facilities, users, operating hours, facility types, facility features, and booking requests efficiently.

Thirdly, the implementation of booking conflict validation minimizes overlapping reservations and improves booking accuracy.

Finally, the responsive user interface developed using Bootstrap provides a consistent user experience across different devices while maintaining system usability.

Overall, the developed Sports Facility Booking Management System fulfills the functional requirements established during the project planning phase.

5.3 System Strengths

The proposed system offers several advantages compared to traditional manual booking methods.

The major strengths include:

- Centralized online booking management.
- Role-based access control for administrators and students.
- Automated booking conflict detection.
- Responsive user interface compatible with multiple devices.
- Secure authentication using Laravel Breeze.
- Organized facility management.
- Interactive booking calendar.
- Improved booking monitoring through dashboard statistics.

- Better data consistency using a relational MySQL database.
- Improved administrative efficiency by reducing manual processes.

These strengths contribute to a more organized and efficient reservation system.

5.4 System Limitations

Although the proposed system successfully fulfills its objectives, several limitations remain.

Firstly, the system is designed only as a web application and does not provide a dedicated mobile application.

Secondly, online payment functionality is not implemented because sports facility reservations within the proposed environment are assumed to be free of charge.

Thirdly, email notifications and SMS reminders are not included in the current implementation.

Finally, the system is intended specifically for Universiti Teknologi MARA and may require customization before deployment in other organizations.

5.5 Future Enhancements

Several improvements may be implemented in future versions of the system to further improve usability and functionality.

Recommended enhancements include:

- Mobile application development for Android and iOS.
- Email notification for booking approval and rejection.
- SMS reminder before booking sessions.
- QR Code check-in for facility usage.
- Integration with university authentication systems.
- Facility image galleries.
- Real-time facility availability updates.
- Advanced dashboard analytics and reporting.
- Export booking data to PDF or Excel.
- Online payment gateway integration if booking fees are introduced.

These enhancements would further improve the effectiveness and usability of the Sports Facility Booking Management System.

5.6 Conclusion

The Sports Facility Booking Management System has been successfully developed using Laravel Framework, MySQL, Bootstrap, and FullCalendar. The system effectively addresses the problems associated with manual sports facility reservation by providing a centralized online platform for both administrators and students.

The implementation of automated booking conflict validation, user management, facility management, and role-based authentication has significantly improved the efficiency of reservation management while minimizing scheduling conflicts.

Overall, the project demonstrates that modern web technologies can be effectively applied to develop a secure, responsive, and user-friendly sports facility reservation system. The completed system satisfies the project objectives and provides a practical solution that can support sports facility management within Universiti Teknologi MARA.

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APPENDICES

Appendix A – Source Code

```php
    // Booking Approval
    Route::patch('/bookings/{booking}/approve', [BookingController::class, 'approve'])
        ->name('bookings.approve');

    Route::patch('/bookings/{booking}/reject', [BookingController::class, 'reject'])
        ->name('bookings.reject');

    Route::patch('/bookings/{booking}/complete', [BookingController::class, 'complete'])
        ->name('bookings.complete');
});
```

Appendix A.1 – Booking Approval Route

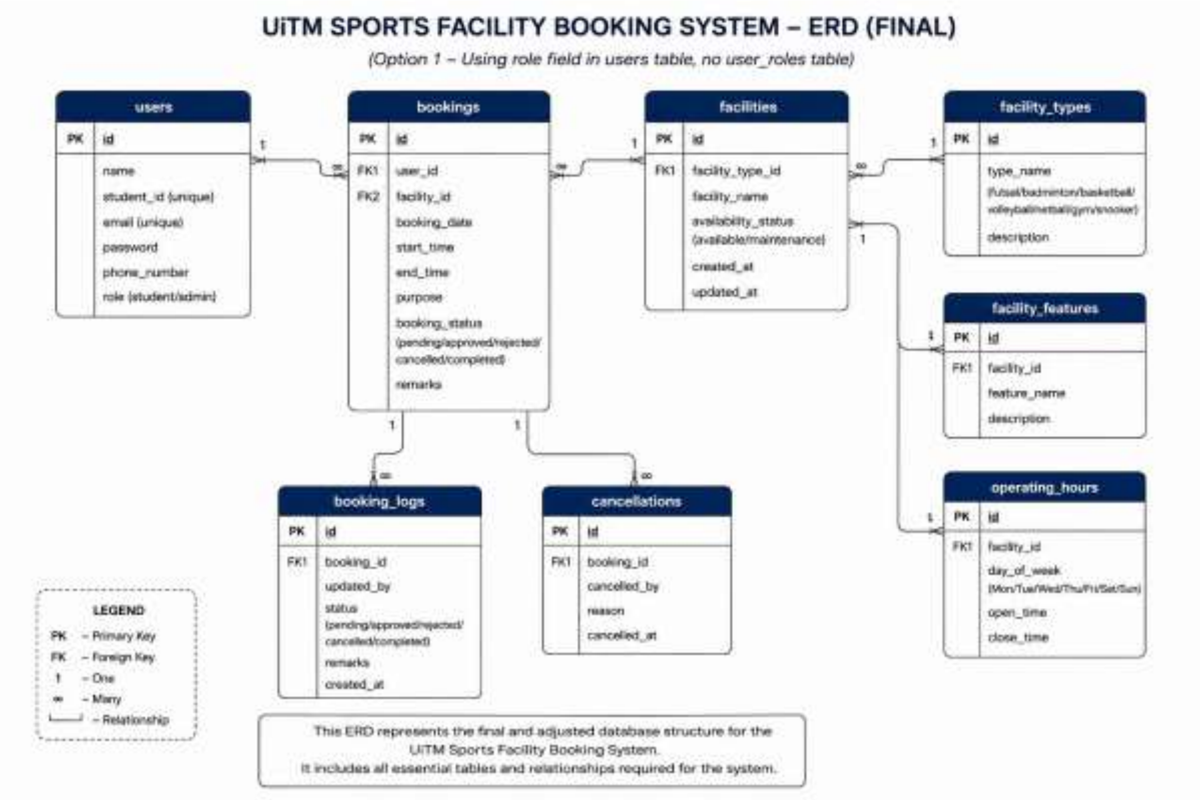
```php
$request->validate([
    'facility_id' => 'required|exists:facilities,id',
    'booking_date' => 'required|date|after_or_equal:today',
    'start_time' => 'required',
    'end_time' => 'required|after:start_time',
    'purpose' => 'required|max:500',
]);
```

Appendix A.2 – Booking Validation

```php
$conflict = Booking::where('facility_id', $booking->facility_id)
    ->where('booking_date', $booking->booking_date)
    ->where('booking_status', 'approved')
    ->where('id', '!=', $booking->id)
    ->where(function ($query) use ($booking) {

        $query->whereBetween('start_time', [
                $booking->start_time,
                $booking->end_time
            ])

            ->orWhereBetween('end_time', [
                $booking->start_time,
                $booking->end_time
            ])

            ->orWhere(function ($q) use ($booking) {

                $q->where('start_time', '<=', $booking->start_time)
                  ->where('end_time', '>=', $booking->end_time);

            });

    })
    ->exists();
```

Appendix A.3 – Booking Conflict Detection

Appendix B – Database Design



Appendix B.1 – Entity Relationship Diagram (ERD)

Table	Action						Rows	Type	Collation	Size	Overhead
bookings	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
booking_logs	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
cache	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
cache_locks	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
cancellations	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
facilities	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
facility_features	Browse	Structure	Search	Insert	Empty	Drop	3	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
facility_types	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
failed_jobs	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
jobs	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
job_batches	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
migrations	Browse	Structure	Search	Insert	Empty	Drop	10	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
operating_hours	Browse	Structure	Search	Insert	Empty	Drop	3	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
password_reset_tokens	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
sessions	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
users	Browse	Structure	Search	Insert	Empty	Drop	3	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
16 tables	Sum						31	InnoDB	utf8mb4_general_ci	528.0 KiB	0 B

Appendix B.2 – Database Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
2	user_id	bigint(20)		UNSIGNED	No	None			Change Drop More
3	facility_id	bigint(20)		UNSIGNED	No	None			Change Drop More
4	booking_date	date			No	None			Change Drop More
5	start_time	time			No	None			Change Drop More
6	end_time	time			No	None			Change Drop More
7	purpose	text	utf8mb4_unicode_ci		No	None			Change Drop More
8	booking_status	enum('pending', 'approved', 'rejected', 'cancelled...	utf8mb4_unicode_ci		No	pending			Change Drop More
9	remarks	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
10	created_at	timestamp			Yes	NULL			Change Drop More
11	updated_at	timestamp			Yes	NULL			Change Drop More

Appendix B.3 – Bookings Table Structure

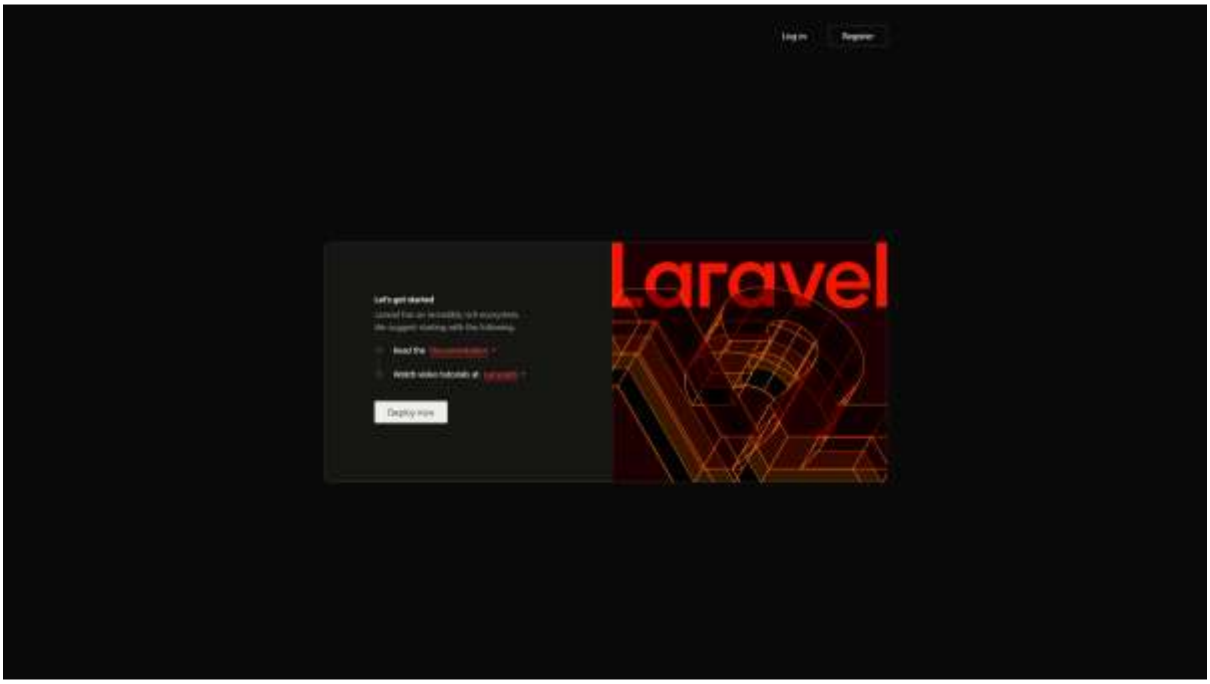
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
2	name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
3	student_id	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
4	email	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
5	email_verified_at	timestamp			Yes	NULL			Change Drop More
6	password	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
7	phone_number	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
8	role	enum('student', 'admin')	utf8mb4_unicode_ci		No	student			Change Drop More
9	remember_token	varchar(100)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
10	created_at	timestamp			Yes	NULL			Change Drop More
11	updated_at	timestamp			Yes	NULL			Change Drop More

Appendix B.4 –Users Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
2	facility_type_id	bigint(20)		UNSIGNED	No	None			Change Drop More
3	facility_name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
4	availability_status	enum('available', 'maintenance')	utf8mb4_unicode_ci		No	available			Change Drop More
5	created_at	timestamp			Yes	NULL			Change Drop More
6	updated_at	timestamp			Yes	NULL			Change Drop More

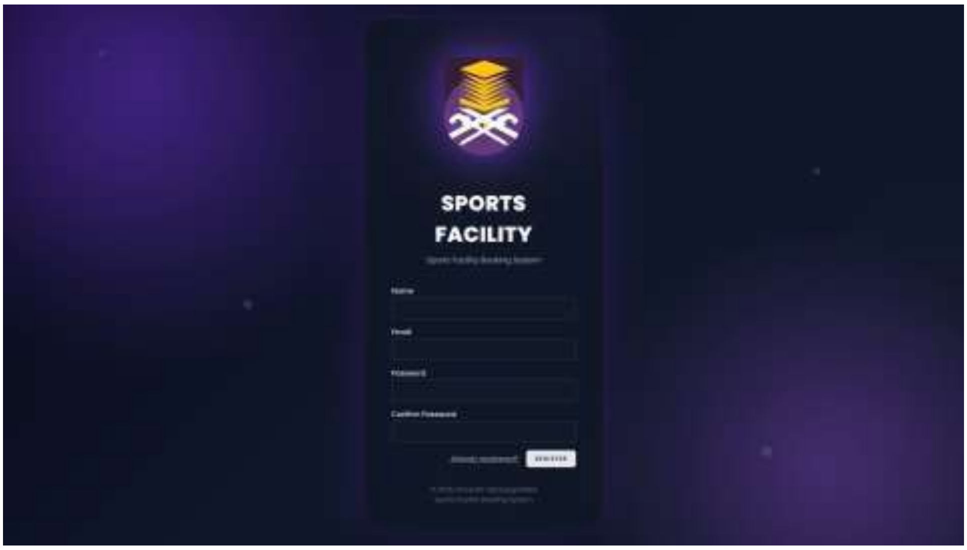
Appendix B.5 – Facilities Table Structure

Appendix C – System Interfaces



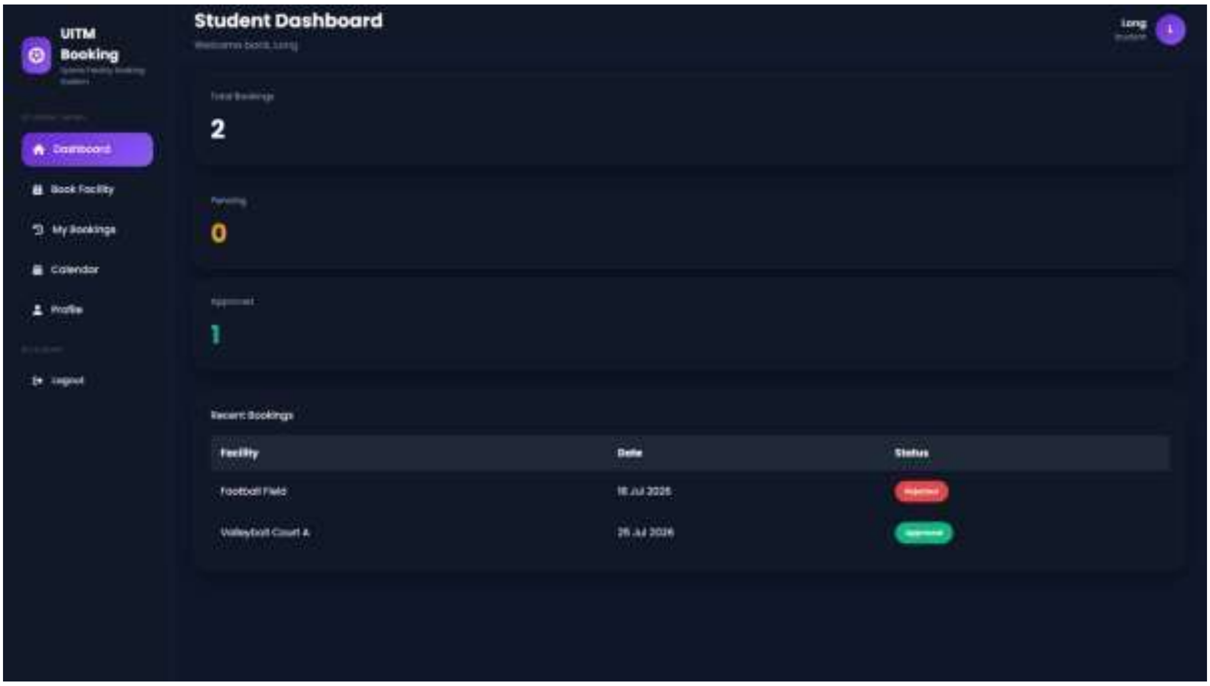
Appendix C.1 – Welcome Page

Appendix C.2 – Login Page

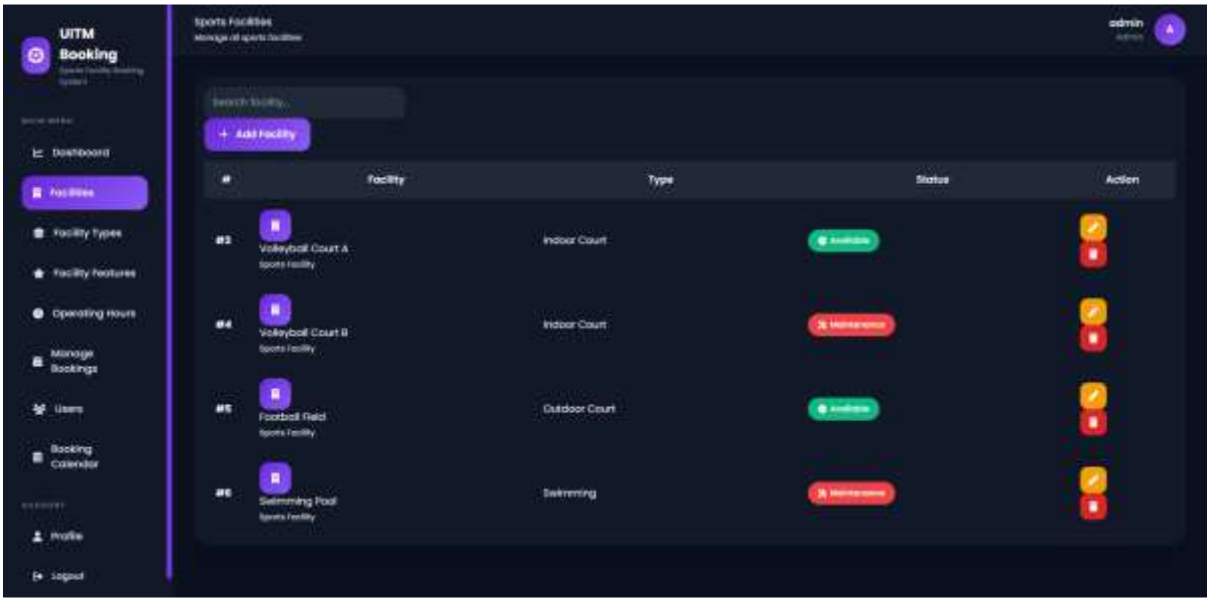


Appendix C.3 – Registration Page

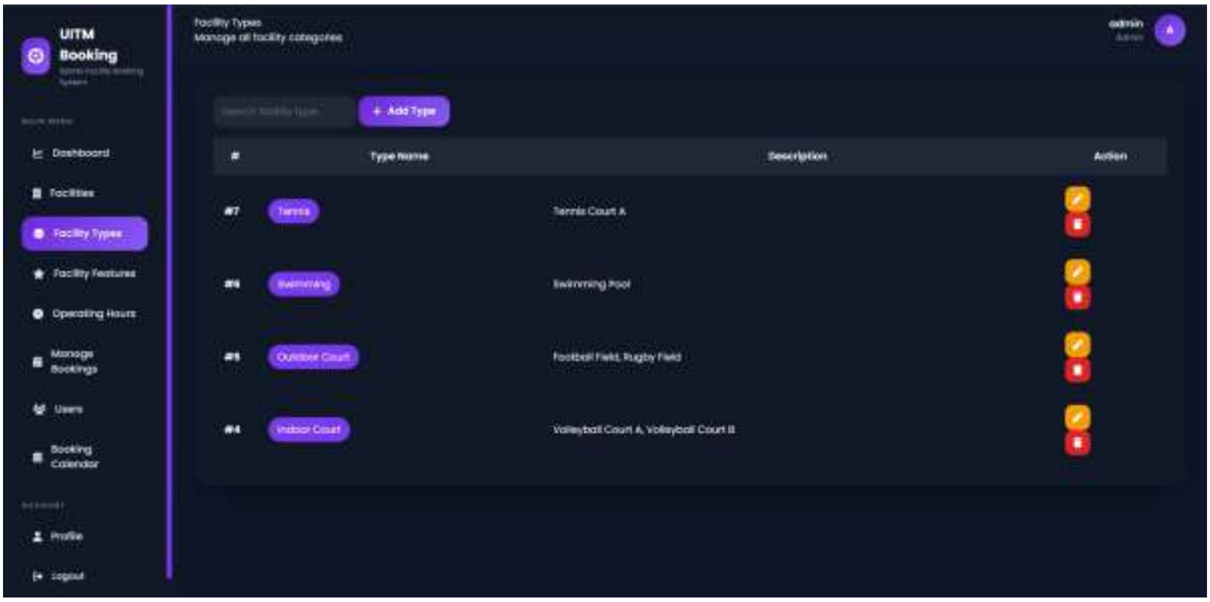
Appendix C.4 – Administrator Dashboard



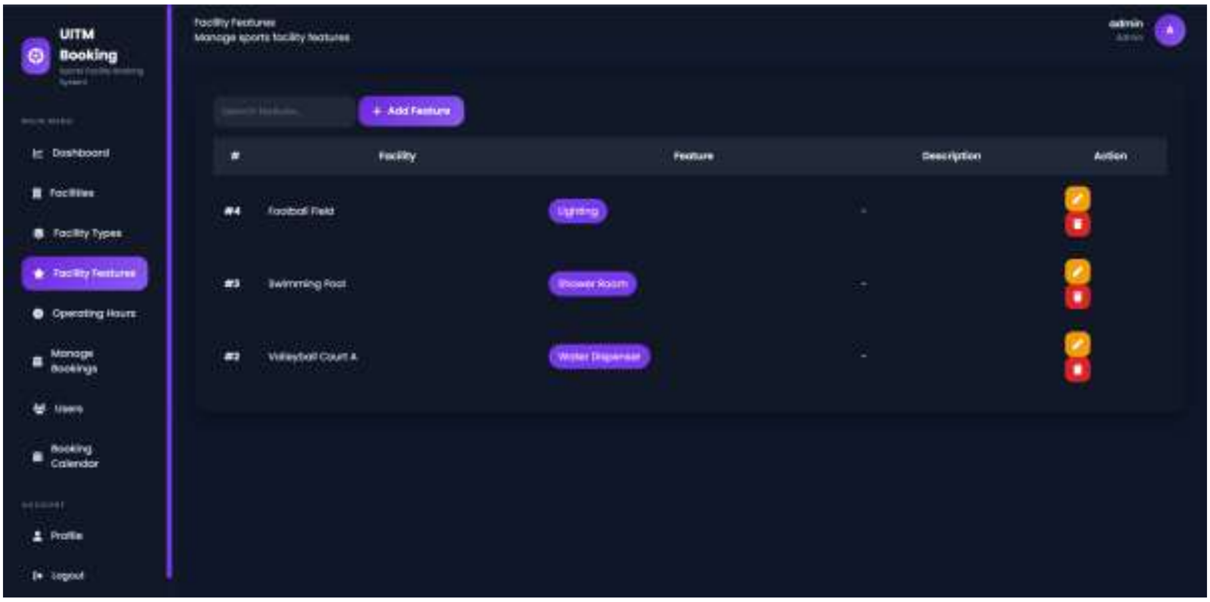
Appendix C.5 – Student Dashboard



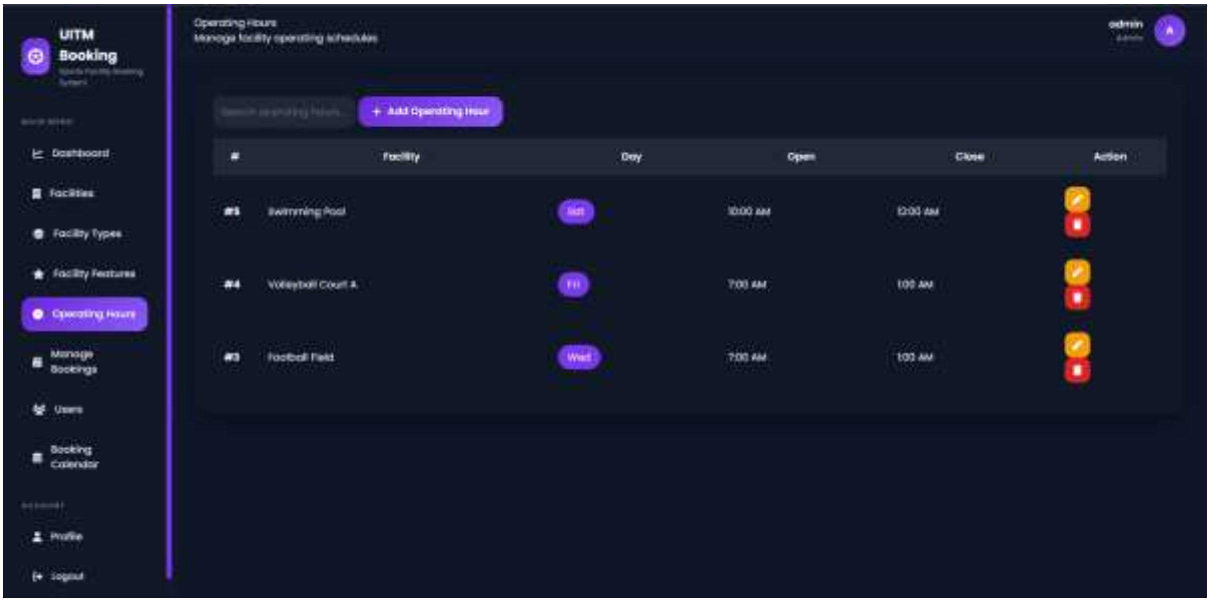
Appendix C.6 – Facility Management



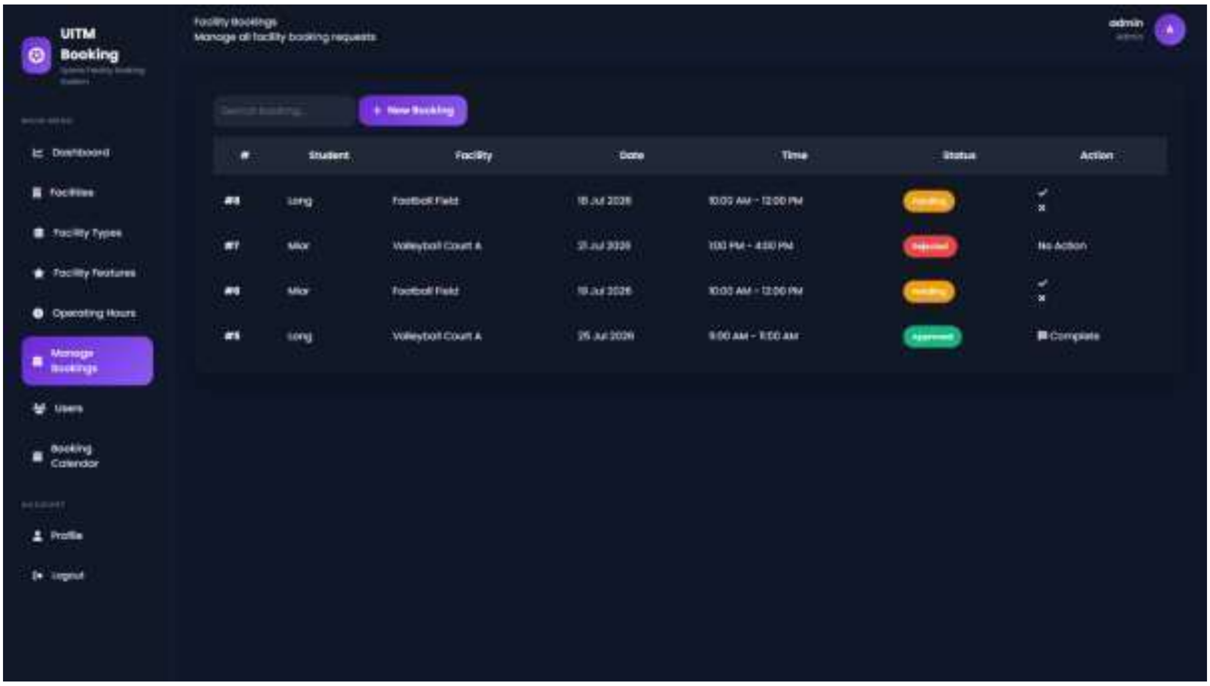
Appendix C.7 – Facility Type Management



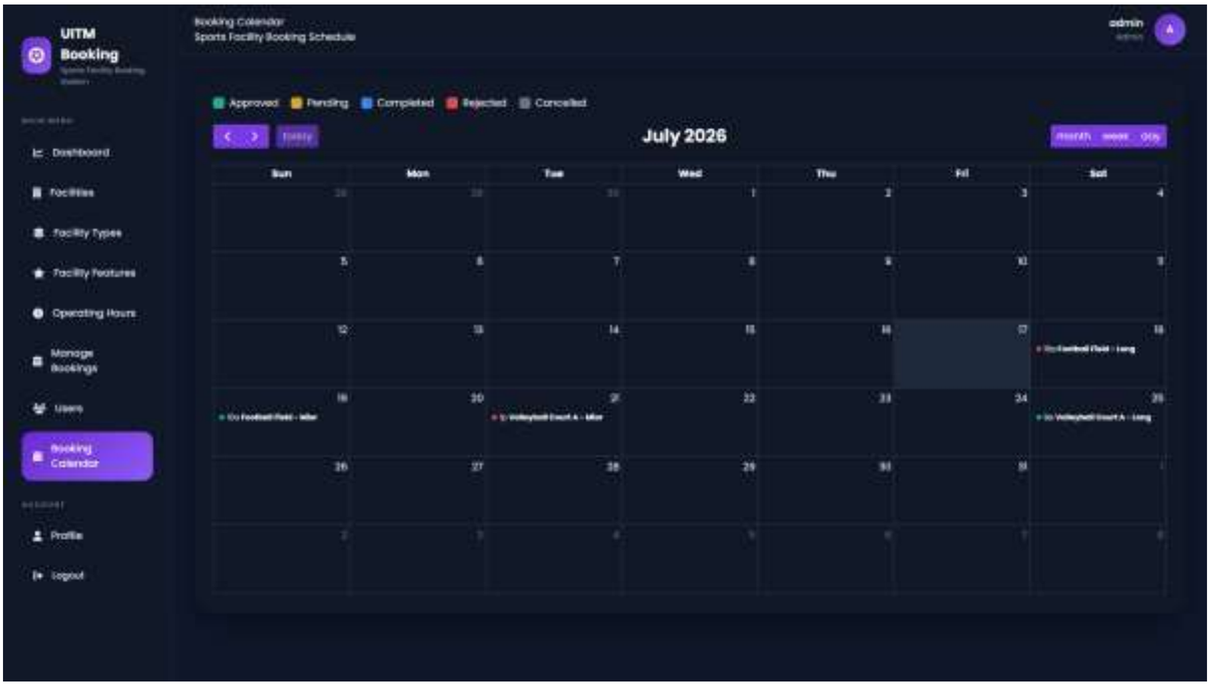
Appendix C.8 – Facility Feature Management



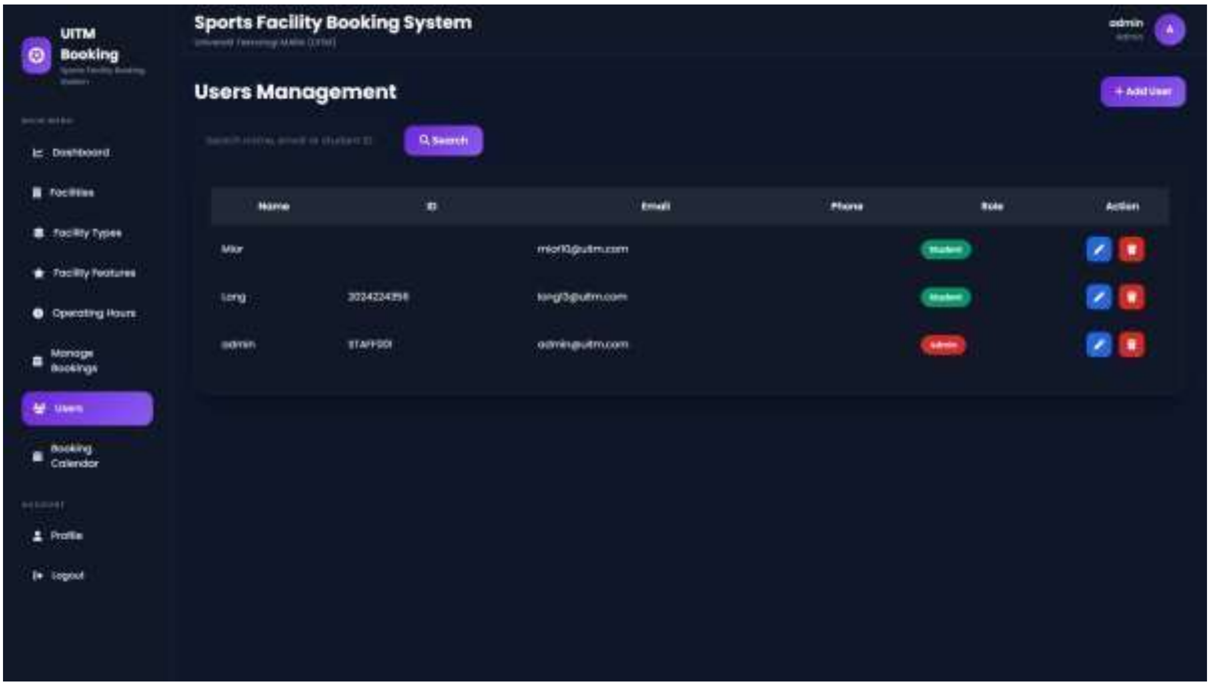
Appendix C.9 – Operating Hours Management



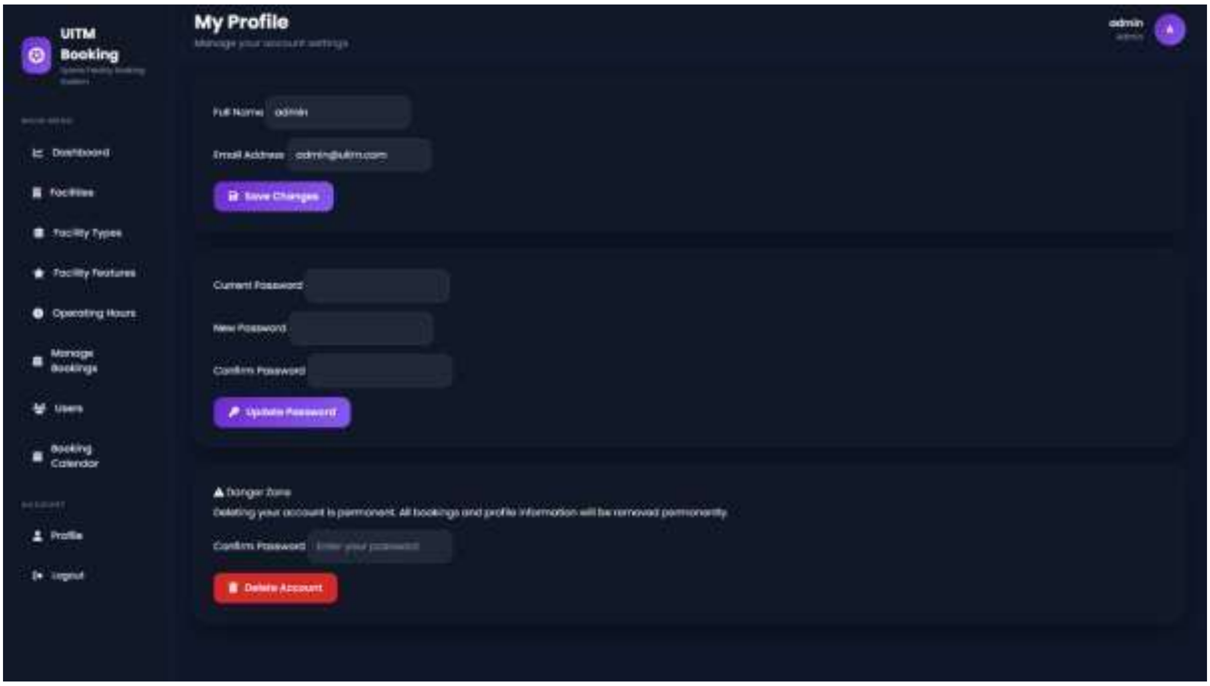
Appendix C.10 – Booking Management



Appendix C.11 – Booking Calendar

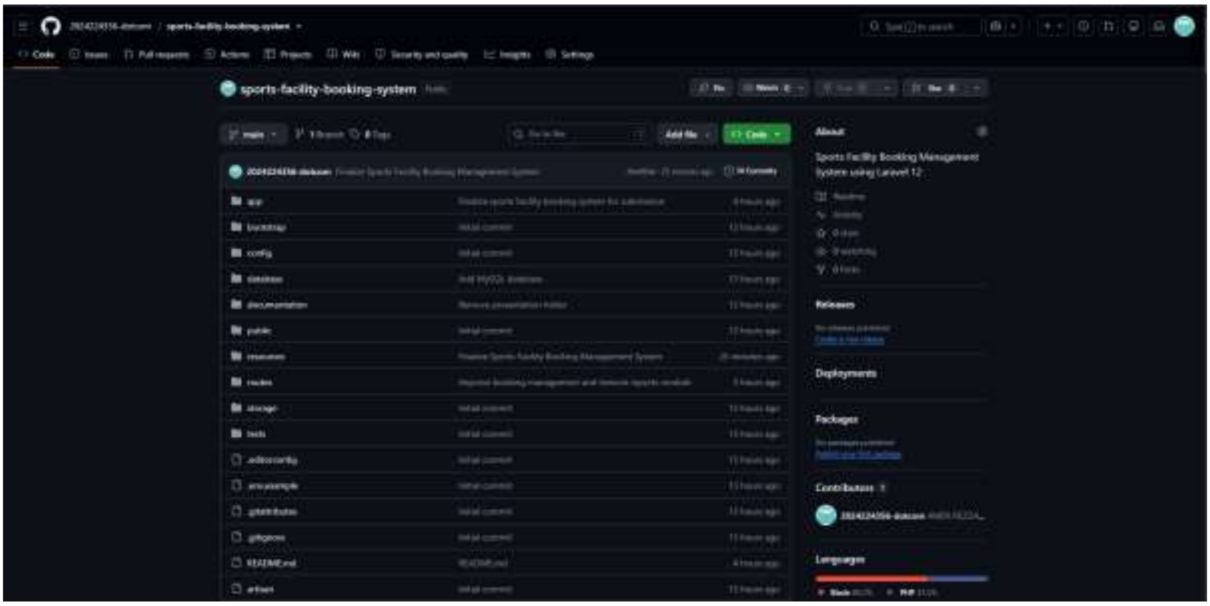


Appendix C.12 – User Management

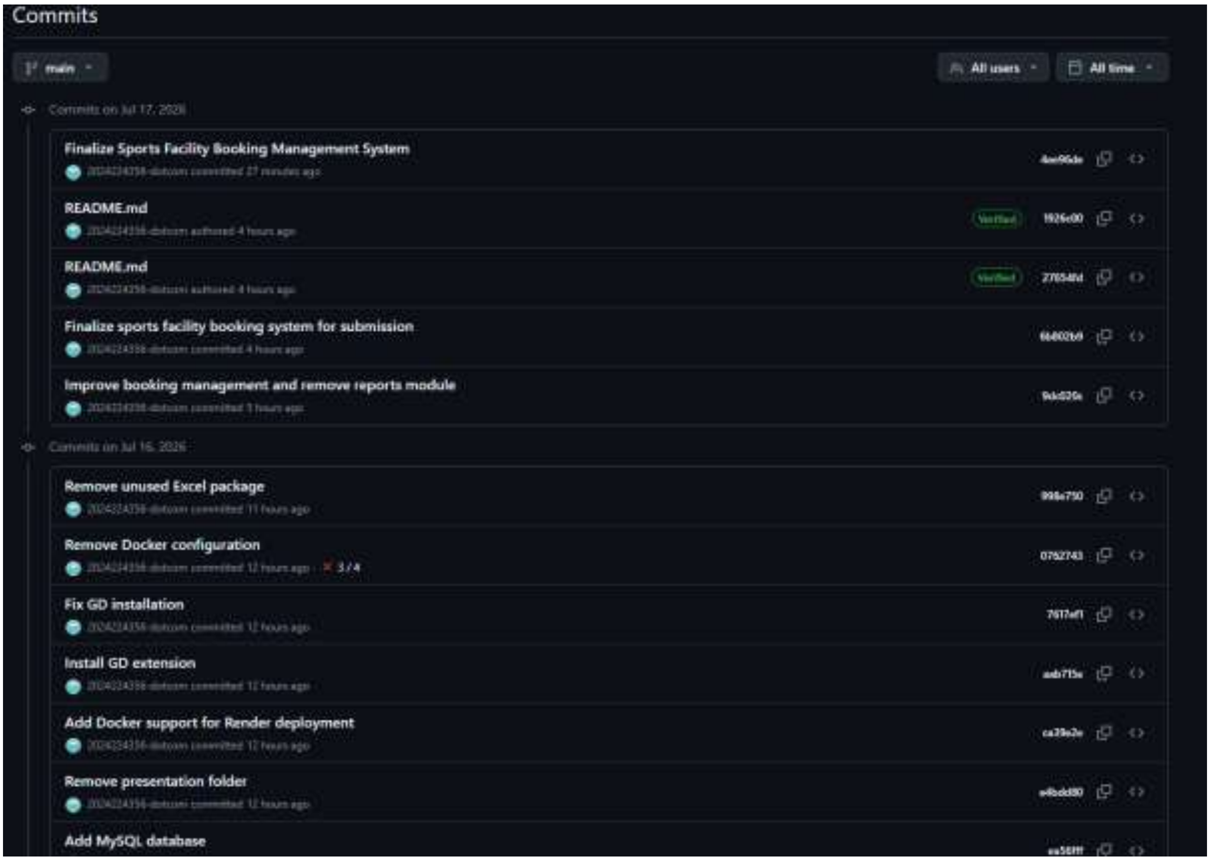


Appendix C.13 – User Profile

Appendix D – Github Repository



Appendix D.1 – GitHub Repository



Appendix D.2 – Latest Commit History